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# Neuromuscular Electrical Stimulation for Children with Dysphagia: A Systematic Review



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|---|---|
| Journal: | *BMJ Open* |
| Manuscript ID | bmjopen-2021-055124 |
| Article Type: | Original research |
| Date Submitted by the Author: | 03-Jul-2021 |
| Complete List of Authors: | Propp, Roni; Tel Aviv University Sackler Faculty of Medicine<br>Gill, Peter; The Hospital for Sick Children, Paediatrics<br>Marcus, Sherna; The Hospital for Sick Children, Rehabilitation<br>Ren, Lily; Stanford Medicine<br>Cohen, Eyal; The Hospital for Sick Children<br>Friedman, Jeremy; The Hospital for Sick Children, Paediatrics<br>Mahant, Sanjay; The Hospital for Sick Children, Paediatrics |
| Keywords: | PAEDIATRICS, Developmental neurology & neurodisability < PAEDIATRICS, NUTRITION & DIETETICS |
| | |



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**Neuromuscular Electrical Stimulation for Children with Dysphagia: A Systematic Review**

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**Word Count**: 3836

**Key Words:** dysphagia, children, systematic review, interventions

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**Abstract**

**Objectives:** Dysphagia in childhood has important health impacts for the child and their family, as well as the healthcare system. However, there is limited information on the effectiveness of interventions to treat dysphagia in children. This systematic review aims to determine the effectiveness of neuromuscular electrical stimulation (NMES) for treatment of oropharyngeal dysphagia in children.

**Methods:** MEDLINE, EMBASE, PsycINFO, CINAHL, CENTRAL and Scopus databases were searched from inception to November 2020. Studies of children (≤18 years) diagnosed with oropharyngeal dysphagia using NMES in the throat/neck region were included. Screening, data extraction, and risk of bias assessment followed the PRISMA guidelines.

**Results:** 10 studies were included (5 RCTs, 4 case series, 1 cohort study; including 393 children, mean or median age below 7 years, including children with neurologic impairments). In all studies, swallowing function improved after NMES treatment. Eight of 10 studies reported on the child's feeding ability, and, with one exception, there was improvement in feeding ability. Few studies reported on health status (N=2), impact on caregiver (N=1), adverse events and harms (N=2), and child's quality of life (N=1). In most studies, outcome follow-up was less than 6 months. The studies demonstrated moderate to high risk of bias.

**Conclusions:** NMES treatment may be beneficial in improving swallowing function for children with dysphagia, however, given the quality of the studies, inadequate outcome reporting, and short follow-up duration, uncertainty remains. Additional well-designed RCTs are needed to establish its effectiveness before its adoption in routine clinical practice.

**Protocol Registration:** Prospero, CRD42019147353



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**Article Summary**

**Strengths and Limitations of this study**

- First systematic review to assess effectiveness of NMES in children with dysphagia.
- Synthesized observational and trial evidence suggests NMES improves swallowing function in children.
- A pooled intervention effect and meta-analysis was not conducted due to clinical heterogeneity.
- High-quality trials are needed before adopting NMES in routine clinical practice.
- Future RCTs should include comprehensive and valid outcome measurement and longer follow-up.

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**Introduction**

Dysphagia which is defined as difficult or abnormal swallowing,[1] occurs in about 1% of children.[2] However, it is more prevalent in specific populations, including premature infants, and children with cerebral palsy, traumatic brain injury, cranio-facial abnormalities, and children with medical complexity.[3] Dysphagia has significant health impacts, including a range of nutritional and respiratory complications.[3,4] Dysphagia can result in poor oral intake, and consequently can lead to malnutrition and failure to thrive. Aspiration and pneumonia are common respiratory complications.[3] Recurrent wheezing and chronic lung disease may also occur.[5] More broadly, dysphagia can impact daily activities and social interaction with peers.[6] It also has an impact on family and caregivers. For example, parents of children with dysphagia experience higher levels of anxiety and stress when compared to parents of children without dysphagia.[7,8] Furthermore, the care of children with dysphagia is costly,[9,10] in part due to increased hospitalization rates, longer hospital lengths of stay, repeated emergency room visits, and increased costs of treatment, including tube feeding.[10] These frequent interactions with the healthcare system can be resource intensive,[9,10] and may place burden and stress on those with dysphagia and their families.[11]

The current management for dysphagia in children includes compensatory strategies[3] and oral motor exercises[12]. Specifically, compensatory strategies include modifying diet (e.g., altering textures, thickening fluids), postural changes, decreasing rate of feeding, and/or using various feeding tools (e.g., different spoon size).[3] These techniques can be helpful in improving the safety of oral feeding in some children but does not address the underlying swallowing issue. In severe cases, permanent tube feeding through a gastrostomy may be required to provide nutrition in a safe way, particularly in children with neurological disabilities.[3]



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A new proposed approach used to treat dysphagia is neuromuscular electrical stimulation (NMES).[13] NMES involves applying electrical current to muscles cutaneously using surface electrodes. Transcutaneous stimulation, administered by occupational therapists and/or speech language pathologists[14], is then combined with traditional swallowing therapy (e.g., compensatory strategies, oral motor exercises, etc.). NMES is usually applied to voluntarily contracting muscles in the neck region, where increased muscle contractions are stimulated through the recruitment of larger and more motor units.[13] Thus, it is hypothesized that NMES improves dysphagia by strengthening muscles involved in swallowing and/or by enhancing the sensory signals of the swallowing response.[13,15] NMES is usually conducted repeatedly over a period of time, for example, some protocols apply NMES up to 5 days a week for a period of 4 weeks, but variation exists.[16]

To date, there is limited evidence of the effectiveness of NMES in the treatment of swallowing dysfunction in children. One of the most cited meta-analyses in adults with dysphagia demonstrated improvement with NMES, and concluded that further research is needed.[17] Several more recent adult studies have also concluded that NMES improves swallowing dysfunction,[18–20] however, high quality studies are still needed.[21] Findings from adult studies should not be extrapolated to children, given that neuroplasticity changes with age and that the etiology of dysphagia in adults is very different (i.e. oropharyngeal malignancies, stroke, dementia). As well, outcomes that are important to children often differ from those that are important to adults. A recent Cochrane systematic review examined the evidence for various interventions to treat dysphagia in children with neurological impairment, but no studies on NMES were included.[22] Therefore, the objectives of this systematic review is to assess the effectiveness of NMES for treatment of oropharyngeal dysphagia (i.e., dysfunction in oral and/or pharyngeal phases of swallowing) in children.



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## Methods

The protocol for this systematic review was registered in PROSPERO (CRD42019147353). The PRISMA guidelines were followed in the development of the protocol and conducting the systematic review. As this study was a systematic review it did not require institutional ethics approval.

### Search Strategy and Information Sources

The search strategy included MeSH headings and free-text terms related to "dysphagia," "neuromuscular electrical stimulation" and "children." The MeSH and free text terms for "children" used a validated filter.[23] MEDLINE (OVID), EMBASE (OVID), PsycINFO (OVID), CINAHL (EBCSO), Cochrane Central Register of Controlled Trials (OVID) and Scopus databases were searched from inception to July 19, 2019 and updated on November 26, 2020. The search strategy was developed and carried out by LR, a reference librarian with expertise in conducting systematic reviews. The complete search strategy for all databases is in Appendix A. The MEDLINE search strategy was translated and adapted for the other databases.

The International Clinical Trials Registry Platform Search Portal and ClinicalTrials.gov were searched. Only completed trials were included in the review, ongoing trials were reported. As well, reference lists of included studies were reviewed. Citation searches were conducted on the included studies. Finally, we contacted experts in the field to identify studies. We used the web-based software platform Covidence to manage records.

### Eligibility Criteria

Studies of children (18 years and younger) diagnosed with oropharyngeal dysphagia were included. A diagnosis of dysphagia using clinical assessment and/or diagnostic testing (e.g., video fluoroscopic



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feeding study [VFSS]) was required; it could not be solely by parental assessment. Studies that included both children and adults were only included if data for children and adults was reported separately and/or could be obtained separately from study authors. Studies of children with esophageal dysphagia were excluded.

All studies using NMES in the throat and neck region were included. Studies using NMES in any other region were excluded. There were no restrictions based on the type of healthcare professional delivering the intervention, setting of the intervention, or number of treatments. Studies using electrical stimulation intramuscularly were excluded.

Randomized controlled trials, cohort studies, case-control studies, cross-sectional studies, and case series were included. Studies published as full-text, abstract only, and unpublished data were included. There were no restrictions by language of publication. For RCTs, studies using standard of care, an alternative intervention, or a placebo treatment as the control group were included. Studies with no control group were also included in the review. Although evidence from RCTs is the gold-standard for assessing effectiveness to establish an intervention for clinical practice, non-RCTs, which are at high risk of bias, were included given the paucity of evidence and to inform future trials.

**Study Selection**

All potentially eligible records were independently screened by title and abstract by two review authors (RP and SM). The full text of studies were retrieved for each included record, and two review authors (RP and SM) independently screened studies for final inclusion. Reasons for excluding studies were recorded. Any disagreements were resolved through discussion and evaluated by a third review author (PG) if necessary.



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**Primary and Secondary Outcomes**

The primary outcome was improvement of swallowing dysfunction as measured by imaging studies, such as videofluoroscopic swallow study (VFSS) and fiberoptic endoscopic evaluation of swallowing (FEES). Secondary outcomes included: 1) swallowing function as reported by a clinician (e.g., a clinical swallowing examination) or by the child and/or their caregiver (e.g. patient reported outcome measures), 2) child's feeding ability (e.g., food type(s) the child can consume, ease of feeding, need for any form of enteral tube feeding, duration of time required for feeding, and child's overall experience of feeding), 3) child's health status (e.g., anthropometric measurements, medical care, and home care), 4) social impact on child (e.g., participation at mealtime), 5) impact on caregiver (e.g., stress associated with feeding), 6) potential adverse events and harms (e.g., aspiration pneumonia and/or recurrent chest infections, mortality), 7) child's quality of life as reported by the child and/or caregiver, and 8) caregiver's quality of life.

**Data Extraction and Management**

One review author (RP) extracted the data and a second review author (SM) verified the extracted data. Data regarding the methods, participants, interventions, outcomes, results was extracted into an a priori developed data extraction form. Any disagreements were resolved through discussion, and if necessary, by involving a third review author (PG). Missing data was obtained by contacting study authors by email, up to a maximum of three email attempts.

**Risk of Bias**

Risk of bias for all included studies was assessed independently by two review authors (RP and SM). Any disagreements were resolved through discussion, and if necessary, by involving a third review author



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(PG). For randomized controlled trial studies, risk of bias was assessed using the Cochrane Collaboration's tool,[24] according to the following domains: random sequence generation, allocation concealment, blinding of participants and researchers, blinding of outcome assessment, incomplete outcome data, selective reporting, and other bias. Each domain was judged as low, high, or unclear risk of bias.[24] For observational studies, risk of bias was assessed using a modified version of the Newcastle-Ottawa assessment tool,[25] according to the following domains: selection, comparability, and outcome. Each domain was then rated using the star system following the coding manual instructions.[25] The scale ranged from 0, indicating the highest risk of bias (i.e., none of the criteria fulfilled), to 7, indicating the lowest risk of bias. The overall risk of bias was summarized across the studies included in the review. The risk of bias for each study was considered when evaluating treatment effects.

**Statistical Analysis**

Randomized controlled trials and observational studies were considered separately. A summary statistic was calculated for the outcomes in each study and grouped according to the pre-defined primary and secondary outcomes. A risk ratio was calculated for dichotomous outcomes. A standardized mean difference was calculated for continuous outcomes, where the numerator was the difference in change scores between the groups and the denominator was the pooled SD of the combined pre and post outcome scores across groups.[26] Median (range) was converted to mean (SD) when necessary. Forest plots of effect estimates were constructed for each study and an overall estimate. Given the heterogeneity between the included studies, a meta-analysis was not conducted, and a pooled intervention effect was not calculated. Review Manager (RevMan) was used to conduct the analyses.

**Patient and Public Involvement**

There was no patient or public involvement in this review.



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**Results**

Of the 1015 unique records screened by title/abstract, 28 full text records were assessed for eligibility, and 17 records met eligibility criteria for the systematic review. An additional 3 articles were included from reference lists and grey literature. Of the 20 records that met eligibility criteria for systematic review, 10 full text articles were included in systematic review (Figure 1) and 10 abstracts were reported on in Appendix B.

**Study Characteristics**

A total of 5 randomized controlled trials (RCT)[27–31], 4 prospective case series (i.e., before and after study with no control group)[32–35], and 1 retrospective cohort study (with a control group)[36] were included in the review (Table 1). One RCT (Gao et al 2018[28]) included NMES in both treatment and control groups, and therefore, was analyzed as a prospective case series. In addition, we were unable to obtain the required data to analyze the study by Christiaanse et al quantitatively,[36] and therefore, this study was analyzed qualitatively. Thus, four studies (4 RCT) were analyzed quantitatively[29–31,37] and 6 studies were analyzed qualitatively.[28,32–36]

The studies were from 6 countries: Egypt,[31] China,[27,28] Canada,[33] Turkey,[29] South Korea,[30,32] and USA.[34–36] The mean age of the included participants was less than 4 years in 8 studies[27–29,31,33–36] and the mean age was 6 years in one study.[30] One study did not specify the age of the participants.[29] A total of 6 studies included participants exclusively with neurological impairment (e.g., cerebral palsy)[27,28,30–33] and 4 studies included some participants with neurological impairment.[29,34–36] The follow up period ranged from a minimum of 4 weeks in 2 studies[29,32] to a maximum of 8 months in one study.[34] The sample size of the studies ranged from 5 participants[32] to 62 participants per group (control and treatment groups)[28]



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(Table 1). Most studies used VFSS before and after the intervention to diagnose dysphagia and evaluate response to treatment. Study funding sources are provided in Appendix C.

**Intervention Characteristics**

Most studies placed the electrodes around the hyoid bone and thyroid notch,[27,28,31,33–36] and in one study, electrode placement was further adjusted on an individual basis.[33] One study placed electrodes on the sublingual muscles,[28] and one study placed electrodes on the submental region.[32] The frequency of sessions ranged from 5 days per week[27–29,32] to 1-2 times per week.[30,31,33–36] The session duration ranged from 20 minutes[27,28,30] to 1 hour,[29,34] and treatment duration ranged from 4 weeks[29,32] to 6 months.[35] In most studies, the voltage applied was 80 Hz and current intensity was adjusted until a therapeutic level was reached.[29,30,32–36] A range of interventions were employed for the control groups, but they all included some form of muscle stimulation (e.g., oral motor exercises)[27–31,35,36] and two studies included a placebo/sham-NMES treatment[30,31] (Table 2).

**Outcomes**

*Swallowing function*

All studies reported on swallowing dysfunction (Table 3 summarizes outcomes reported across studies), and in all cases, swallowing function improved over the course of NMES treatment. However, only five used imaging to define swallowing function, and therefore, we grouped swallowing dysfunction as measured by imaging or clinical exam for this outcome. The standardized mean difference for improvement of swallowing dysfunction in treatment compared to control groups in the RCTs ranged from 0.18 (95% CI -0.7, 1.06)[30] to 1.49 (95% CI 0.57, 2.41)[29] (Figure 2). The study by Lv et al 2019[27] (RCT) exhibited a graded improvement in swallowing function with increasing current intensity in the different treatment groups (Figure 2). In two small prospective case series studies, all participants improved in



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swallowing function (Online Table 4). In the study by Ma et al 2019[32] (prospective case series study) and Gao et al 2018[28] (RCT treated as cohort study), the scales used to grade swallowing dysfunction showed an overall improvement (Online Table 4). Specifically, in the Gao et al 2018 study, 71% of participants improved their swallowing effectiveness. Improvement in the swallowing activities ranged from 13% improvement on the chewing movement subscale to 19% on the movement of lips and tongue subscale.[28] In the Ma et al 2019 study, there was a 45% improvement in swallowing semisolids and 33% improvement in swallowing liquids as measured by the Penetration Aspiration Scale (PAS).[32] In the Andreoli et al 2019 study, there was a statistically significant improvement in mean swallowing function as measured by the Functional Oral Intake Scale (FOIS) (from 3.07 [1.94] to 4.47 [2.26]). In the Christiaanse et al 2011 study, there was no significant difference observed between the treatment and control groups for change in swallowing function measured by FOIS(Table C).[35]

*Child's feeding ability*

Eight of 10 studies reported on child's feeding ability.[27–30,33–36] In all but one study, there was improvement in feeding ability over the course of the NMES treatment (Figure 2, Table C). Similarly, the study by Lv et al 2019 exhibited a graded improvement in feeding ability with increasing current intensity in different treatment groups (Figure 2).[27] Marcus et al 2019 reported that all participants improved in their ability to swallow different consistencies safely and 5 of 5 participants progressed from tube feeding to full or partial oral feeding.[33] Similarly, all participants in the Rice et al 2012 study improved in their ability to swallow different consistencies.[34] There was an improvement in the feeding ability of 79% of participants in the Gao et al 2018 study.[28] In the Andreoli et al 2019 study, improvement was noted in 7 of 8 patients with gastrostomy dependence; for patients without gastrostomy dependence, full oral feeds without restriction was achieved in 6 of 7 patients (Online Table 4).[35] The standardized mean difference for improvement in feeding behaviours compared with the



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control group in the study by Song et al 2015[30] (RCT) was 0.39 (95% CI -0.49, 1.28). The exception was the study by Christiaanse et al 2011[36] (retrospective cohort study), which found that the intervention was not associated with an improvement in feeding ability compared with the control group (Online Table 4).

*Child's health status*

Lv et al 2019[27] reported on nutritional status, which improved over the course of treatment with standardized mean differences ranging from 0.59 (95% CI -0.15, 1.32) to 1.33 (95% CI 0.53, 2.13) for the different treatment groups compared with the control group using increasing voltage of NMES. El-Sheikah et al 2020[31] found that weight increased over the course of treatment (standardized mean difference of 0.04 (95% CI -0.58, 0.66) (Figure 2).

*Potential adverse events and harms*

Marcus et al 2019[33] reported on adverse events and harms, which included mild skin irritation at the placement of electrode and resolved without treatment (N = 6). Andreoli et al 2019[35] reported that the NMES was well tolerated and there were no treatment complications. Adverse events were not reported in the other studies.

*Child's quality of life as reported by the child*

Gao et al 2018[28] reported on child's quality of life using the Gessell scale and there was an overall improvement over the course of treatment. More specifically, there was a mean improvement of 15% for the adaptabilities subscale, 11% for the language behaviour subscale, and 12% for the personal social development subscale (Online Table 4).



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*Impact on caregiver*

Marcus et al 2019[33] reported on the impact on the caregiver, specifically 5 of 7 caregivers felt that their child's feeding improved and were satisfied with the intervention. However, a common reported negative aspect was the frequent need to travel to the hospital for treatments.

No studies reported on the social impact on child or caregiver's quality of life (Table 3).

**Quality of Studies**

The risk of bias for RCTs were assessed using the Cochrane collaboration tool.[24] All studies were rated low for attrition bias and reported bias, most studies had a high level of bias for areas related to blinding, most studies were low for random sequence generation, and all studies were unclear for allocation concealment (Online Table 5). Furthermore, for all the RCTs, there was no reported primary outcome or sample size estimation. The observational studies were assessed for risk of bias using a modified version of the Newcastle-Ottawa scale[25]: 3 of 5 studies scored 5 (out of 7),[28,33,34] one study scored 1 (out of 7)[32], one study score 5 (out of 7),[35] and one study scored 6 (out of 8)[36] (Online Table 6).

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**Discussion**

Dysphagia is a serious, often chronic health problem with significant impacts for the child, family, and the healthcare system.[3,4,6–8] It can lead to poor oral intake and malnutrition, aspiration and pneumonia, as well as impact daily activities and social interaction with peers.[3,4,6] Parents and caregivers of children with dysphagia experience increased anxiety and stress,[7,8] and there are increased costs and utilization of the healthcare system.[9,10] Currently, there is no effective treatment that targets the underlying swallowing disorder in children with dysphagia. NMES is a new proposed intervention for the treatment of dysphagia. In this review, we analyzed 10 studies, including 5 randomized controlled trials,[27–31] 4 prospective case series studies without a control group,[32–35] and 1 retrospective cohort study[36] where NMES was used to treat dysphagia in children. Overall, both the RCTs and observational studies reported an improvement in swallowing dysfunction following NMES intervention, which demonstrates that NMES treatment may be beneficial for children with dysphagia. However, given the limitations in the primary studies, well-designed trials are needed before adopting this intervention in clinical practice.

This is the first systematic review to investigate the use of NMES for children with dysphagia. Several systematic reviews and meta-analysis have been conducted in different populations of adults with dysphagia (e.g., stroke, head and neck cancer), with some reported benefits on swallowing dysfunction,[18–20] however, these results need to be interpreted with caution in children. Two recent reviews examined the management and interventions for children with dysphagia,[22,38] but no studies on NMES were included.

To date, the studies in this area are limited in number, and many demonstrated moderate to high risk of bias. For example, most RCTs did not blind participants, personnel, and outcome assessment. Though studies randomized patients to treatment and control groups, there was insufficient information



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provided about allocation concealment across all studies. In the Song et al study,[30] there was a significant difference between treatment and control groups at baseline. The observational studies had similar issues: the representativeness of the sample was unclear in two studies,[32,36] there was a significant difference between treatment and control groups in one study,[36] and one study provided insufficient data on participants.[32] Lastly, in 7 of 10 studies, the follow up period was less than 6 months.[27–33] A longer follow-up period (e.g. 1 year) is important to determine whether any benefit derived from NMES is temporary or sustained. Together, these factors contribute to moderate to high risk of bias in the included studies.

This review has several limitations. First, due to clinical heterogeneity, we did not calculate a pooled intervention effect and conduct a meta-analysis. Second, we included a wide range of studies, several of which did not have a control group. In these cases, it cannot be determined that the observed effect was due to the intervention; however, when there is limited evidence, it is important to survey the literature more broadly. There were 10 abstracts and unpublished studies that were not included in the analysis, however, there was insufficient data reported to include these reports in our review.

We have several recommendations for future trials. We recommend the use of imaging studies to objectively evaluate swallowing function, such as VFSS and FEES, which are established imaging techniques used to evaluate swallowing function.[39,40] Furthermore, it is imperative to use validated outcome measures for children with dysphagia that can be applied across different pediatric populations. In terms of validated patient and/or parent reported outcomes, there are several potential measures,[41–44] however, there is no established standard that would allow for systematic reporting and comparison across studies. Second, only several of the a priori primary and secondary outcomes were reported across the studies. Dysphagia has impacts across biological, social, and psychological realms,



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and therefore, it is important for future trials to assess a broad range of outcomes to develop a more complete clinical picture. For example, none of the studies in this review reported on social impact on the child, impact on the caregiver, and the caregiver's quality of life. In order to ensure outcomes important to patients and their families are assessed, we recommend patients and their families be included as partners when designing future studies. In addition, all trials should be reported according to the CONSORT guidelines, including sample size justification as well as blinding participants and personnel, which were notably missing in many of these studies. As well, the effects of the intervention should be assessed in different patient populations (e.g., young infants and older children) and aetiologies (e.g., premature infants, neurologic impairment). It would be important to assess outcomes for a sufficiently long duration of follow up to determine if the effect of NMES persists over time. However, in most of these studies, the follow up period was less than 6 months.

**Conclusions**

Dysphagia in childhood has important health impacts for the child and their family, as well as the healthcare system. Evidence from small trials with moderate to high risk of bias and observational studies suggest that NMES improves swallowing function in children. There is limited evidence on other important outcomes. Further high-quality trials are needed before NMES is adopted in routine clinical practice.

**Acknowledgements**

**Competing Interests:** There are no competing interests to disclose.

**Funding/Support:** This study was supported by a grant from the Pediatric Consultants Partnership, The Hospital for Sick Children, Toronto, Canada.



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**Role of Funder/Sponsor:** The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

**Authors Contributions:** Ms. Roni Propp conceptualized and designed the study, collected data, carried out the analyses and interpretation of data, and drafted and revised the manuscript. Drs. Peter Gill, Eyal Cohen, Jeremy Friedman and Ms. Sherna Marcus conceptualized and designed the study, interpretated the data, and reviewed and revised the manuscript. Ms. Lily Ren conceptualized and designed the study, designed and executed the search strategy, and reviewed and revised the manuscript. Dr. Sanjay Mahant conceptualized and designed the study, collected data, analysed and interpreted the data, drafted the initial manuscript, and reviewed and revised the manuscript. All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

**Data Statement:** Not applicable.

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**Table 1. Study Characteristics**

| Identification | | | Number of participants | | Characteristics of participants | | | | Follow Up Time Length |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Treatment | | Control | | |
| Author | Year | Country | Treatment | Control | **Age,** Years Mean (SD) | **Neurological impairment,** N (%) | **Age,** Years Mean (SD) | **Neurological impairment,** N (%) | |
| **Randomized controlled trials** | | | | | | | | | |
| El-Sheikh[31] | 2020 | Egypt | 20 | 20 | 3.26 (1.16) | 20 (100%) | 3.96 (0.98) | 20 (100%) | 2 months |
| Lv[27] | 2019 | China | 45* | 15 | 3.93 (0.73) | 45 (100%) | 3.92 (0.85) | 15 (100%) | 3 months |
| Gao[28]** | 2018 | China | 62 | 62 | 3.01 (1.12) | 62 (100%) | 3.02 (1.02) | 62 (100%) | 3 months |
| Serel Arslan[29] | 2017 | Turkey | 12 | 12 | 1.01 (0.63) | 1 (8.3%) | 1.28 (0.93) | 1 (8.3%) | 1 month |
| Song[30] | 2015 | South Korea | 10 | 10 | 6.2 (2.78) | 10 (100%) | 6.00 (2.40) | 10 (100%) | 2 months |
| **Prospective case series** | | | | | | | | | |
| Andreoli[35] | 2019 | USA | 15 | --- | 2.51 (3.20) | 11 (73%) | --- | --- | 6 months |
| Ma[32] | 2019 | South Korea | 5 | --- | Not reported | 5 (100%) | --- | --- | 1 month |
| Marcus[33] | 2019 | Canada | 7 | --- | 0.66 (0.37) | 7 (100%) | --- | --- | 2-4 months |
| Rice[34] | 2012 | USA | 5 | --- | 1.45 (1.05) | 2 (40%) | --- | --- | 3-8 months |
| **Retrospective cohort studies** | | | | | | | | | |
| Christiaanse[36] | 2011 | USA | 47 | 46 | 2.58 (1.66) | 32 (68%) | 0.83 (0.68) | 16 (35%) | 6 months |

*3 treatment groups; 15 participants in each group

**Included NMES in both treatment and control groups, and therefore, was analyzed as a prospective case series.



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**Table 2. NMES and Control Group Intervention Details**

| Author | Year | Additional intervention components | Electrode placements | Frequency of sessions | Duration of NMES session | Duration of treatment | Voltage applied | Current intensity | Control Intervention |
|---|---|---|---|---|---|---|---|---|---|
| **Randomized controlled trials** | | | | | | | | | |
| El-Sheikh[31] | 2020 | Oral motor exercises | (1) over the neck between hyoid and jaw (2) side of face | Twice weekly | 20 minutes | 2 months | 60 Hz | 7mA to 15mA | Oral motor exercises; placebo NMES |
| Lv[27] | 2019 | --- | (1) above hyoid bone, (2) above the thyroid upper notch | 5 days a week | 20 minutes | 3 months | 80Hz | 10mA, 15mA, 20mA (3 treatment groups) | Routine rehabilitation training (oral massage and acupuncture) |
| Gao[28]* | 2018 | Oral motor exercises, Acupuncture | Superficial muscles and sublingual muscles | 5 days a week | 20 minutes | 3 months | 20-30 Hz | 3-7 mA | Oral motor exercises, NMES |
| Serel Arslan[29] | 2017 | Thermal tactile application (TTA), Hyolaryngeal mobilization | (1) below the jaw, (2) above the thyroid notch | 5 days a week | 30 minutes | 1 month | 80 Hz | Until a therapeutic level was reached | Non-nutritive stimulations |
| Song[30] | 2015 | Oral sensorimotor treatment | (1) over the throat between jaw and hyoid (2) between hyoid and thyroid notch | Twice weekly | 20 minutes | 2 months | 80 Hz | Typical level ranged from 3mA to 5 mA (determined by palpation for muscle contraction) | Oral sensorimotor treatments and sham-NMES |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Prospective case series studies** | | | | | | | | | |
| Andreoli[35] | 2019 | Comprehensive feeding therapy program | (1) superior aspect of the thyroid cartilage, (2) along tongue base superior to hyoid bone | Weekly | Up to 50 minutes or as tolerated | 6 months | 80 Hz | 7.5 (2.7) mAmp (maximal mean (SD)) | No control group |
| Ma[32] | 2019 | --- | Submental region | 5 days a week | 30 minutes per day | 1 month | 80Hz | Until the patients felt a minimal stimulation level | No control group |
| Marcus[33] | 2019 | Feeding to elicit swallowing during NMES session | (1) above the hyoid, or (2) around the thyroid notch Location determined based on the child's swallowing impairment | Twice a week for 2 months; Further 2 months for those in whom improvement was not observed at 2 months | 40 minutes (median; range 20-45) | 2 to 4 months | 80 Hz | Range from 3 to 16 mA, based on the response of the child | No control group |
| Rice[34] | 2012 | Oral motor stimulation techniques between sessions | (1) above the thyroid notch, or (2) around the thyroid notch | 1-2 times a week | 1 hour | Varied | 80 Hz | 0 to 25mA (adjusted for each child to their highest tolerated level) | No control group |
| Retrospective cohort studies | | | | | | | | | |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Christiaanse[36] | 2011 | Non-nutritive oral motor stimulation or a meal (during NMES); Oral motor exercises between sessions | (1) below the jaw, (2) above the thyroid notch<br><br>In general, infants would only have a pair of electrodes overlying the anterior belly of digastric muscles | 2.9 times a week (mean, range 0.7-4.6) | 30-45 minutes depending on patient age | 2.2 months (mean, range 0.5-6.2 months) | 80Hz | Adjusted until a therapeutic level was reached | Usual diet manipulation, oral motor therapy |

*Included NMES in both treatment and control groups, and therefore, was analyzed as a prospective case series.

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**Table 3. Summary of Outcomes Reported**

| Outcome | Study | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | **El-Sheikh 2020**[31] | **Andreoli 2019**[35] | **Lv 2019**[27] | **Gao 2018**[28] | **Serel Arslan 2017**[29] | **Song 2015**[30] | **Ma 2019**[32] | **Marcus 2019**[33] | **Rice 2012**[34] | **Christiaanse 2011** |
| Swallowing function | | | | | | | | | | |
| Measured by imaging studies | - | - | - | - | + | - | + | + | + | + |
| Reported by clinician | - | + | + | + | - | + | - | - | - | - |
| Reported by child and/or their caregiver | + | - | - | - | - | - | - | - | - | - |
| Child's feeding ability | - | + | + | + | + | + | - | + | + | + |
| Child's health status | + | - | + | - | - | - | - | - | - | - |
| Social impact on child | - | - | - | - | - | - | - | - | - | - |
| Impact on caregiver | - | - | - | - | - | - | - | + | - | - |
| Potential adverse events and harms | - | - | - | - | - | - | - | + | - | - |
| Child's quality of life | - | - | - | + | - | - | - | - | - | - |
| Caregiver's quality of life | - | - | - | - | - | - | - | - | - | - |

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**Figure Legends**

**Figure 1. PRISMA Flowchart**

**Figure 2. Outcomes in Randomized Controlled Trials.**[1] [1]Studies did not report all outcomes; for standardized mean difference, point estimate of 0 = no effect, and for risk ratio, point estimate of 1 = no effect. Abbreviations, RCT, randomized controlled trial.

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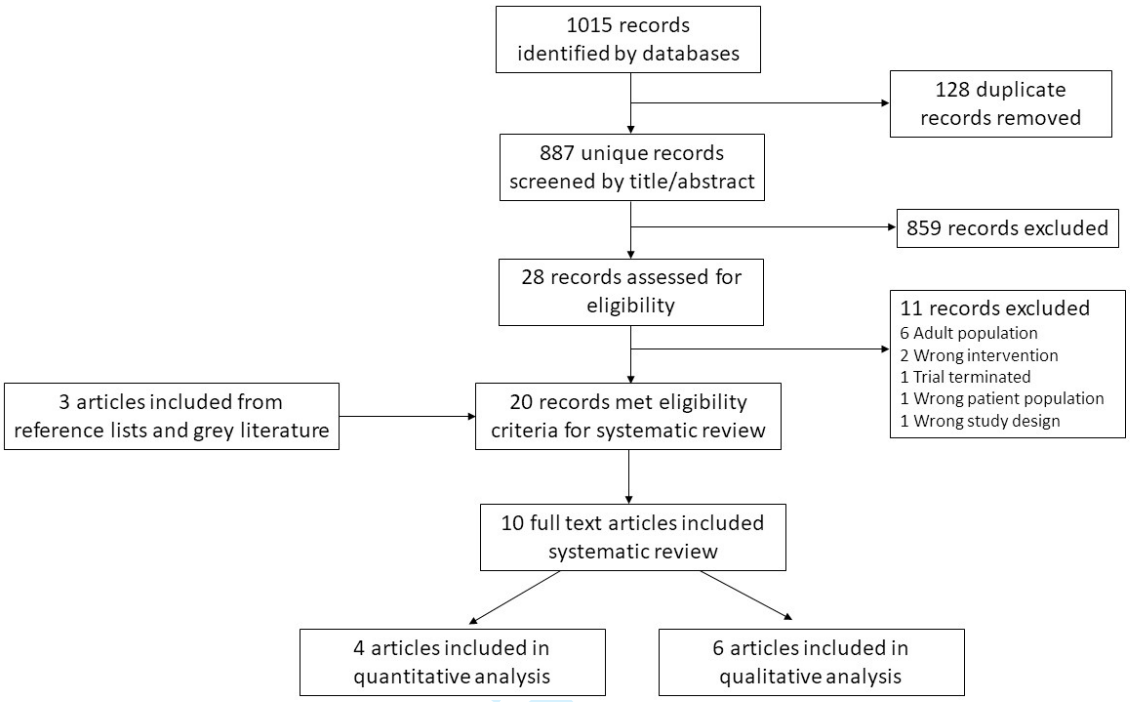
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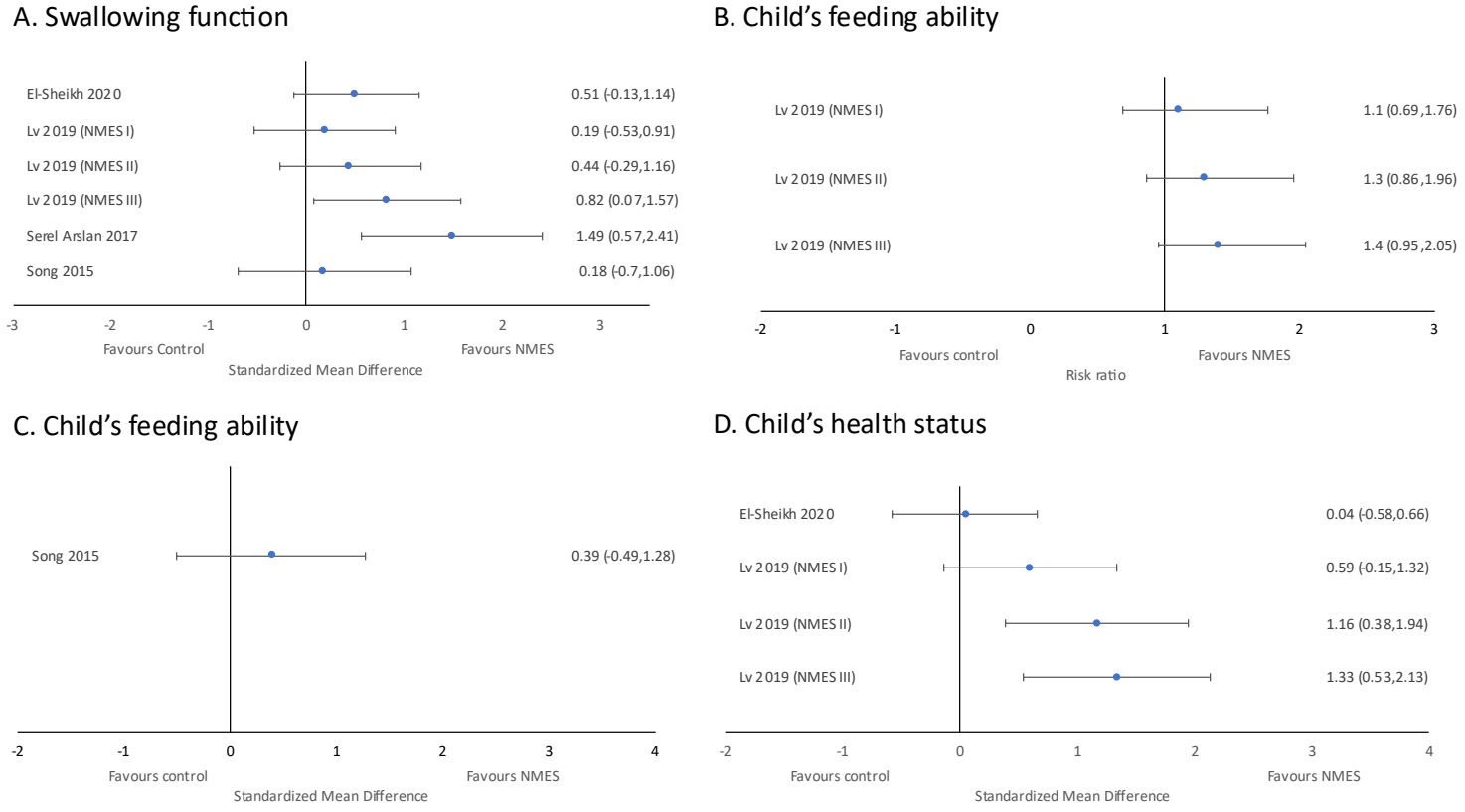
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**Figure 1. PRISMA Flowchart**




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**Figure 2. Outcomes in Randomized Controlled Trials**[1]




[1]Studies did not report all outcomes; for standardized mean difference, point estimate of 0 = no effect, and for risk ratio, point estimate of 1 = no effect. Abbreviations, RCT, randomized controlled trial.

**Online Figures and Tables**

**Online Table 4. Outcomes in Observational Studies**[1]

| Study | Number of Participants | Outcome | Scale | Description |
|---|---|---|---|---|
| Andreoli 2019 | 15 | Swallowing function | Functional Oral Intake Scale[2] | (Mean (SD)): Pre = 3.07 (1.94), Post = 4.47 (2.26)<br>Of those with aspiration and penetration 10/12 improved and of those with aspiration 7/10 resolved |
| | | Feeding ability | Number of participants | For patients with gastrostomy dependence, improvement was noted in 7/8 patients; for patients without gastrostomy dependence, full oral feeds without restriction was achieved in 6/7 patients |
| Christiaanse 2011[36] | 93 | Swallowing function | Functional Oral Intake Scale[2] | Mean change (SD): Treatment = 0.77 (1.22), Control = 0.39 (0.98), however, this difference was not statistically significant |
| | | Feeding ability | Number of participants | Treatment group: 80% of patients were fed by the same method before and after the intervention; 4 participants were no longer using enteral tubes at 6 month follow-up<br>Control group: 70% of patients were fed by the same method before and after the intervention; 7 participants were no longer using oral tubes at 6 month follow-up |
| Ma 2019[32] | 5 | Swallowing function | Penetration Aspiration Scale[3] | Semisolid (Mean (SD)): Pre = 3.8 (1.5), Post = 2.1 (1.2)<br>Liquid (Mean (SD)): Pre = 6.4 (2.1), Post = 4.3 (2.5) |
| Marcus 2019[33] | 7 | Swallowing function | Number of participants | All participants improved |
| | | Child's feeding ability | Number of participants | 5 of 7 participants improved<br>Pre: 5 not safe for oral feeding (tube fed)<br>Post: 3 full oral feeding, 2 partial oral feeding, 0 tube fed |
| | | Impact on caregiver | Number of participants | 5 of 7 caregivers felt feeding improved and were satisfied with the intervention; frequent need to travel to the hospital for treatments |
| Gao 2018[28] | 124 | Child's feeding ability | TDS Classification | 88 of 124 participants improved |
| | | | Efficacy | 98 of 124 participants had an effective or significantly effective response to intervention |
| | | Child's quality of life | Gessell | Adaptabilities: 15% improvement |

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| | | | | |
|---|---|---|---|---|
| | | Swallowing function | Number of participants[4] | Language behaviour: 11% improvement<br>Personal social development: 12% improvement<br>Movement of lips and tongue: 24 of 124 participants improved<br>Chewing movement: 16 of 124 participants improved<br>Swallowing function: 18 of 124 participants improved<br>Free eating: 20 of 124 participants improved |
| Rice 2012[34] | 5 | Swallowing function<br><br>Child's quality of life | Number of participants<br><br>Number of participants | All participants improved; Improved swallow safety and ability to advance in consistencies<br>All participants improved |

[1]Only studies without a control group were included in this table
[2]Higher scores indicate improvement
[3]Lower scores indicates improvement
[4]Number of participants in denominator is participants who exhibited this activity before treatment

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**Online Table 5. Risk of Bias Assessment for the Included Observational Studies Using the Newcastle Ottawa Scale (NOS)[1]**

| Author | Year | Control Group | Selection (3/4 items)[2] | Comparability (1 item) | Outcome (3 items) | Score (Max = 7/8)[3] |
|---|---|---|---|---|---|---|
| Andreoli[35] | 2019 | No | ★ ★ ★ | ★ | ★★ | 5 |
| Ma[32] | 2019 | No | ★ | | | 1 |
| Marcus[33] | 2019 | No | ★ ★ ★ | ★ | | 4 |
| Gao[28 4] | 2018 | No | ★ ★ | ★ | ★ | 4 |
| Rice[34] | 2012 | No | ★ ★ | ★ | ★ | 4 |
| Christiaanse[36] | 2011 | Yes | ★ ★ ★ ★ | | ★ ★ | 6 |

[1]A modified version of the Newcastle Ottawa scale was used (Appendix C).

[2]For prospective case series (with no control group), selection consists of 3 items. For retrospective studies (with control group), selection consists of 4 items.

[3]For prospective case series (with no control group), scores ranged from 0 (highest risk of bias) to 7 (lowest risk of bias). For retrospective studies (with control group), scores ranged from 0 (highest risk of bias) to 8 (lowest risk of bias).

[4]This study included NMES in both treatment and control groups, and therefore, was analyzed as a prospective case series.

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**Online Table 6. Risk of Bias Assessment for the Included Randomized Controlled Trials Using the Cochrane Collaboration Tool**[1]

| Author | Year | Selection bias | | Performance bias | | Detection bias | Attrition bias | Reporting bias | Other bias | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Random sequence generation | Allocation concealment | Blinding of participants | Blinding of personnel | Blinding of outcome assessment | Incomplete outcome data | Selective reporting | Similarity at baseline | Follow up |
| El-Sheikh[31] | 2020 | ? | ? | + | ? | ? | ? | + | + | - |
| Lv[27] | 2019 | + | ? | - | - | - | + | + | + | - |
| Serel Arslan[29] | 2017 | + | ? | - | - | ? | + | + | - | - |
| Song[30] | 2015 | ? | ? | + | - | - | + | + | + | - |

[1] + indicates low risk of bias, - indicates high risk of bias, ? indicates unclear.

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**Appendix A. Search Strategies for electronic databases**

**MEDLINE (Ovid) search strategy**

1. deglutition disorders/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw,kf.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw,kf.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw,kf.
5. (feeding* or regurgit*).tw,kf.
6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw,kf.
7. or/1-6
8. oropharynx/ or oropharyn*.tw,kf.
9. (dysphag* or disorder* or dysfunc* or impair*).tw,kf.
10. 8 and 9
11. 7 or 10
12. electric stimulation therapy/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw,kf.
14. 12 or 13
15. exp child/ or exp "congenital, hereditary, and neonatal diseases and abnormalities"/ or exp infant/ or adolescent/ or exp pediatrics/ or child, abandoned/ or exp child, exceptional/ or child, orphaned/ or child, unwanted/ or minors/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre-term or preterm* or premature birth* or NICU or preschool* or pre-school* or kindergarten* or kindergarden* or elementary school* or nursery school* or (day care* not adult*) or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or pre-pubesc* or prepubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

**Embase (Ovid) search strategy**

1. dysphagia/ or swallowing/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw,kw.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw,kw.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw,kw.
5. (feeding* or regurgit*).tw,kw.
6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw,kw.
7. or/1-6
8. oropharynx/ or oropharyn*.tw,kw.
9. (dysphag* or disorder* or dysfunc* or impair*).tw,kw.
10. 8 and 9
11. 7 or 10
12. electrotherapy/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw,kw.

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14. 12 or 13
15. juvenile/ or exp adolescent/ or exp child/ or exp postnatal development/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre-term or preterm* or premature birth* or NICU or preschool* or pre-school* or kindergarten* or kindergarden* or elementary school* or nursery school* or (day care* not adult*) or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or pre-pubesc* or prepubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

**PsycINFO (Ovid) search strategy**

1. dysphagia/ or swallowing/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw,id.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw,id.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw,id.
5. (feeding* or regurgit*).tw,id.
6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw,id.
7. or/1-6
8. oropharyn*.tw,id.
9. (dysphag* or disorder* or dysfunc* or impair*).tw,id.
10. 8 and 9
11. 7 or 10
12. electrical stimulation/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw,id.
14. 12 or 13
15. adolescent development/ or childhood development/ or pediatrics/ or exp Congenital Disorders/ or child characteristics/ or child abuse/ or exp child welfare/ or chronically ill children/ or child neglect/ or child psychiatry/ or child psychopathology/ or exp child care/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre term or preterm* or premature birth or NICU or preschool* or pre school* or kindergarten* or elementary school* or nursery school* or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or prepubesc* or pre pubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

**Cochrane Central Register of Controlled Trials (OVID) search strategy**

1. deglutition disorders/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw.
5. (feeding* or regurgit*).tw.

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6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw.
7. or/1-6
8. oropharynx/ or oropharyn*.tw.
9. (dysphag* or disorder* or dysfunc* or impair*).tw.
10. 8 and 9
11. 7 or 10
12. electric stimulation therapy/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw.
14. 12 or 13
15. exp child/ or exp "congenital, hereditary, and neonatal diseases and abnormalities"/ or exp infant/ or adolescent/ or exp pediatrics/ or child, abandoned/ or exp child, exceptional/ or child, orphaned/ or child, unwanted/ or minors/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre-term or preterm* or premature birth* or NICU or preschool* or pre-school* or kindergarten* or kindergarden* or elementary school* or nursery school* or (day care* not adult*) or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or pre-pubesc* or prepubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

**CINAHL (EBSCO) search strategy**

1. (MH "deglutition disorders") OR TX (dysphagia* OR deglutition OR swallow OR swallows OR swallowing OR feeding* OR odynophagia* OR feeding* OR regurgit*) OR TX (deglut* N3 (abnormal* OR disorder* OR dysfunc* OR impair*)) OR TX (oropharynx* OR trachea* OR lung* OR pulmon* ) N3 aspirat* ) ) OR TX (pharyng* N3 (dysphag* OR dysfunct* OR disorder* OR impair*))
2. ((MH "oropharynx") OR TX (oropharyn*)) AND TX (dysphag* OR disorder* OR dysfunc* OR impair*)
3. 1 or 2
4. (MH "electric stimulation therapy")
5. TX (electrostimulation or NMES or "electrical stimulation" or "electric stimulation" or electrotherapy or "electro therapy")
6. 4 or 5
7. (MH "Child+") OR (MH "Child Behavior+") OR (MH "Child Abuse, Sexual") OR (MH "Child Behavior Disorders+") OR (MH "Child, Medically Fragile") OR (MH "Child Day Care") OR (MH "Child Passenger Safety") OR (MH "Child Abuse+") OR (MH "Child Custody") OR (MH "Child Behavior Checklist") OR (MH "Child Nutritional Physiology+")
8. (MH "Child Development Disorders, Pervasive+") or (MH "Adolescence+") or (MH "Minors (Legal)") or (MH "Child Mortality")
9. SO (child* or pediatric* or paediatric* or adolescent)
10. TX (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or "pre-term" or preterm or "premature birth*" or NICU or preschool* or "preschool*"

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or kindergarten* or "elementary school*" or "nursery school*" or schoolchild* or toddler* or boy or boys or girl* or "middle school*" or pubescen* or juvenile* or teen* or youth* or "high school*" or adolesc* or prepubesc* or "pre-pubesc*")

11. or/7-10
12. 3 and 6 and 11

**Scopus search strategy**

( ( TITLE-ABS-KEY ( dysphagia* OR deglutition OR swallow OR swallows OR swallowing OR feeding* OR odynophagia* OR feeding* OR regurgit* ) ) OR ( TITLE-ABS-KEY ( deglut* W/3 ( abnormal* OR disorder* OR dysfunc* OR impair* ) ) ) OR ( TITLE-ABS-KEY ( ( oropharynx* OR trachea* OR lung* OR pulmon* ) W/3 aspirat* ) ) OR ( TITLE-ABS-KEY ( pharyng* W/3 ( dysphag* OR dysfunct* OR disorder* OR impair* ) ) ) OR ( TITLE-ABS-KEY ( oropharyn* ) AND TITLE-ABS-KEY ( dysphag* OR disorder* OR dysfunc* OR impair* ) ) ) AND ( ( TITLE-ABS-KEY ( electrostimulation OR nmes OR "electrical stimulation" OR "electric stimulation" OR electrotherapy OR "electro therapy" ) ) AND ( TITLE-ABS-KEY ( pediatric* OR paediatric* OR child* OR newborn* OR congenital* OR infan* OR baby OR babies OR neonat* OR "pre term" OR preterm* OR "premature birth" OR nicu OR preschool* OR "pre school*" OR kindergarten* OR "elementary school*" OR "nursery school*" OR schoolchild* OR toddler* OR boy OR boys OR girl* OR "middle school*" OR pubescen* OR juvenile* OR teen* OR youth* OR "high school*" OR adolesc* OR prepubesc* OR "pre pubesc*" ) ) )

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**Appendix B. Citation information for unpublished studies and abstracts**

| Author | Year | Title | Journal |
|---|---|---|---|
| Biernacka | 2019 | Evaluation of physical and emotional well-being of children undergoing electrostimulation of swallowing | J Pediatr Gastroenterol Nutr |
| Winnicka | 2019 | Improving swallowing function due to electrical stimulation in group of children-preliminary results from an innovative method of treatment | J Pediatr Gastroenterol Nutr |
| Winnicka | 2019 | Improvement of oral feeding due to electrical stimulation in children with primary dysphagia | J Pediatr Gastroenterol Nutr |
| Radika | 2018 | Role of NMES in dysphagia management for children with hypertonia and dystonia: a case study | Dysphagia |
| Biernacka | 2017 | Evaluation of physical and emotional comfort of children undergoing electrostimulation of swallowing- Initial report | J Pediatr Gastroenterol Nutr |
| Andreoli | 2016 | Neuromuscular stimulation and feeding in medically complex children | Otolaryngol Head and Neck Surg |
| Demir | 2015 | Swallowing function of a pierre robin sequence with absence epiglottis: A case report | Dysphagia |
| Mitchell | 2010 | The effectiveness of NMES (VitalStim) therapy in the neonatal population | Unpublished |
| Christiaanse | 2005 | Experience with pharyngeal electrical stimulation: A new treatment modality for the management of pediatric dysphagia | Dev Med Child Neurol |
| Chatburn | 2000 | Electrical stimulation for treatment of dysphagia in children failing conventional therapy | Respiratory Care |

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**Appendix C. Study Funding Source**

| Study | Year | Funding Source |
|---|---|---|
| Andreoli[35] | 2019 | Not indicated |
| Christiaanse[36] | 2011 | No financial support |
| El-Sheikh[31] | 2020 | Not indicated |
| Gao[28] | 2018 | Not indicated |
| Lv[27] | 2019 | National Natural Science Foundation of China Early Career Fund, Young Talents of Science, Education and Health in Jiangsu Province |
| Ma[32] | 2019 | Not indicated |
| Marcus[33] | 2019 | Peer-reviewed grant from the Department of Paediatrics, The Hospital for Sick Children, Toronto |
| Rice[34] | 2012 | Not indicated |
| Serel Arslan[29] | 2017 | No financial support |
| Song[30] | 2015 | No financial support |

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | 4-5 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 5 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | 6 |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 6-7 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 6 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | 6 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 6-7 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 6 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 6-9 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 8 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | 9 |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | 9 |

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 8 |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | 9 |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | 10 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | 10, Table 1 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | 14, Online table 5 and 6 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | 11-15, Figure 2 |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | NA |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 14, Online table 5 and 6 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | NA |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 15 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 16 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 17 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 17-18 |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

# Neuromuscular Electrical Stimulation for Children with Dysphagia: A Systematic Review



| | |
|---|---|
| Journal: | *BMJ Open* |
| Manuscript ID | bmjopen-2021-055124.R1 |
| Article Type: | Original research |
| Date Submitted by the Author: | 13-Dec-2021 |
| Complete List of Authors: | Propp, Roni; Tel Aviv University Sackler Faculty of Medicine<br>Gill, Peter; The Hospital for Sick Children, Paediatrics<br>Marcus, Sherna; The Hospital for Sick Children, Rehabilitation<br>Ren, Lily; Stanford Medicine<br>Cohen, Eyal; The Hospital for Sick Children<br>Friedman, Jeremy; The Hospital for Sick Children, Paediatrics<br>Mahant, Sanjay; The Hospital for Sick Children, Paediatrics |
| Primary Subject Heading: | Paediatrics |
| Secondary Subject Heading: | Rehabilitation medicine, Neurology |
| Keywords: | PAEDIATRICS, Developmental neurology & neurodisability < PAEDIATRICS, NUTRITION & DIETETICS |
| | |



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# Neuromuscular Electrical Stimulation for Children with Dysphagia: A Systematic Review

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**Word Count**: 4148

**Key Words:** dysphagia, children, systematic review, interventions

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## Abstract

**Objectives:** Dysphagia in childhood has important health impacts for the child and their family, as well as the healthcare system. This systematic review aims to determine the effectiveness of neuromuscular electrical stimulation (NMES) for treatment of oropharyngeal dysphagia in children.

**Methods:** A search was performed on November 2020 in MEDLINE (from 1946), EMBASE (from 1947), PsycINFO (from 1806), CINAHL (from 1937), CENTRAL (from 1996) and Scopus (from 1970) databases. Studies of children (≤18 years) diagnosed with oropharyngeal dysphagia using NMES in the throat/neck region were included. Screening, data extraction, and risk of bias assessment followed the PRISMA guidelines. Risk of bias was assessed using the Cochrane Collaboration's tool for RCTs and a modified Newcastle-Ottawa assessment for observational studies. A meta-analysis was not conducted due to clinical heterogeneity in studies.

**Results:** 10 studies were included (5 RCTs, 4 case series, 1 cohort study; including 393 children, mean or median age below 7 years, including children with neurologic impairments). In all studies, swallowing function improved after NMES treatment. The SMD for improvement of swallowing dysfunction in treatment compared to control groups in the RCTs ranged from 0.18 (95% CI -0.7, 1.06) to 1.49 (95% CI 0.57, 2.41). Eight of 10 studies reported on the child's feeding ability, and, with one exception, there was improvement in feeding ability. Few studies reported on health status (N=2), impact on caregiver (N=1), adverse events and harms (N=2), and child's quality of life (N=1). In most studies, outcome follow-up was less than 6 months. The studies demonstrated moderate to high risk of bias.

**Conclusions:** NMES treatment may be beneficial in improving swallowing function for children with dysphagia, however, given the quality of the studies, inadequate outcome reporting, and short follow-up duration, uncertainty remains. Well-designed RCTs are needed to establish its effectiveness before its adoption in clinical practice.

**Protocol Registration:** Prospero, CRD42019147353




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## Article Summary

### Strengths and Limitations of this study

- First systematic review to assess effectiveness of NMES in children with dysphagia.
- Synthesized observational and trial evidence suggests NMES improves swallowing function in children.
- A pooled intervention effect and meta-analysis was not conducted due to clinical heterogeneity.
- High-quality trials are needed before adopting NMES in routine clinical practice.
- Future RCTs should include comprehensive and valid outcome measurement and longer follow-up.

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**Introduction**

Dysphagia is defined as difficult or abnormal swallowing.[1] A United States national health survey conducted in 2012 found that amongst 61 million children about 1% suffered a swallowing problem lasting greater than 1 week in the previous year.[2] Dysphagia is prevalent in specific populations, including premature infants, and children with cerebral palsy, traumatic brain injury, cranio-facial abnormalities, and children with medical complexity.[1] The prevalence of dysphagia ranges from 58%-99% in children with cerebral palsy[3] and 10.5% in premature infants (<37 weeks).[1] Dysphagia has significant health impacts, including a range of nutritional and respiratory complications.[4,5] Dysphagia can result in poor oral intake, and consequently can lead to malnutrition and failure to thrive. Aspiration and pneumonia are common respiratory complications.[1] Recurrent wheezing and chronic lung disease may also occur.[6] More broadly, dysphagia can impact daily activities and social interaction with peers.[7] It also has an impact on family and caregivers. For example, parents of children with dysphagia experience higher levels of anxiety and stress when compared to parents of children without dysphagia.[8,9] Furthermore, the care of children with dysphagia is costly,[10,11] in part due to increased hospitalization rates, longer hospital lengths of stay, repeated emergency room visits, and increased costs of treatment, including tube feeding.[11] These frequent interactions with the healthcare system can be resource intensive,[10,11] and may place burden and stress on those with dysphagia and their families.[12]

The current management for dysphagia in children includes compensatory strategies[4] and therapeutic interventions.[13] Specifically, compensatory strategies include modifying diet (e.g., altering textures, thickening fluids, changing taste or temperature), postural changes, pacing, using various feeding tools (e.g., different spoon size), and/or changing environments.[4,14] Therapeutic interventions include oral motor exercises, which involve lip control and tongue control, and can be completed with the aid of a therapist and different equipment. For some children, despite compensatory strategies and therapeutic



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interventions, permanent tube feeding through a gastrostomy may be required to provide nutrition in a safe way, particularly in children with neurological disabilities.[4]

A proposed approach used to treat dysphagia is neuromuscular electrical stimulation (NMES).[15] NMES involves applying electrical current to muscles cutaneously using surface electrodes. Transcutaneous stimulation, administered by occupational therapists and/or speech language pathologists[16], is then combined with traditional swallowing therapy (e.g., compensatory strategies, therapeutic interventions, etc.). NMES is usually applied to voluntarily contracting muscles in the neck region, where increased muscle contractions are stimulated through the recruitment of larger and more motor units.[15] Thus, it is hypothesized that NMES improves dysphagia by strengthening muscles involved in swallowing and/or by enhancing the sensory signals of the swallowing response.[15,17] NMES is usually conducted repeatedly over a period of time, for example, some protocols apply NMES up to 5 days a week for a period of 4 weeks, but variation exists.[18]

Evidence that supports NMES use in routine practice is uncertain. One of the most cited meta-analyses in adults with dysphagia demonstrated improvement with NMES, and concluded that further research is needed.[19] Several more recent adult studies have also concluded that NMES improves swallowing dysfunction,[20–22] however, high quality studies are still needed.[23] Findings from adult studies should not be extrapolated to children, given that the etiology of dysphagia in adults is very different (i.e. oropharyngeal malignancies, stroke, dementia) and that the goal of rehabilitation is different in adults (i.e., adults are undergoing rehabilitation for lost skills and children are undergoing rehabilitation to develop skills). In addition, neuroplasticity changes with age. As well, outcomes that are important to children often differ from those that are important to adults. A recent Cochrane systematic review examined the evidence for various interventions to treat dysphagia in children with neurological



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impairment, but no studies on NMES were included.[24] Therefore, the objective of this systematic review is to assess the effectiveness of NMES for treatment of oropharyngeal dysphagia in children.

## Methods

The protocol for this systematic review was registered in PROSPERO (CRD42019147353). The PRISMA guidelines were followed in the development of the protocol and conducting the systematic review. Of note, several study authors who were involved in the systematic review (SM, JF, SM), have conducted research in this area and previously conducted a study on NMES. As this study was a systematic review it did not require institutional ethics approval.

### Search Strategy and Information Sources

The search strategy included MeSH headings and free-text terms related to "dysphagia," "neuromuscular electrical stimulation" and "children." The MeSH and free text terms for "children" used a validated filter.[25] MEDLINE (OVID from 1946), EMBASE (OVID from 1947), PsycINFO (OVID from 1806), CINAHL (EBCSO from 1937), Cochrane Central Register of Controlled Trials (OVID from 1996) and Scopus (from 1970) databases were searched from inception to July 19, 2019 and updated on November 26, 2020. The search strategy was developed and carried out by LR, a reference librarian with expertise in conducting systematic reviews. RP, SM, PG, SM were also involved in the development of the search terms. Three of the five authors are clinicians involved in the care of children with dysphagia (SM, SM, PG). The complete search strategy for all databases is in Appendix A. The MEDLINE search strategy was translated and adapted for the other databases.

The International Clinical Trials Registry Platform Search Portal and ClinicalTrials.gov were searched. Only completed trials were included in the review, ongoing trials were reported. As well, reference lists



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of included studies were reviewed. Citation searches were conducted on the included studies. Finally, we contacted experts in the field to identify studies. We used the web-based software platform Covidence to manage records.

**Eligibility Criteria**

Studies of children (18 years and younger) diagnosed with oropharyngeal dysphagia were included. A diagnosis of dysphagia using clinical assessment and/or diagnostic testing (e.g., video fluoroscopic feeding study [VFSS]) was required; it could not be solely by parental assessment. Studies that included both children and adults were only included if data for children and adults was reported separately and/or could be obtained separately from study authors. Studies of children with esophageal dysphagia were excluded.

All studies using NMES in the throat and neck region were included. Studies using NMES in any other region were excluded. There were no restrictions based on the type of healthcare professional delivering the intervention, setting of the intervention, or number of treatments. Studies using electrical stimulation intramuscularly were excluded.

Randomized controlled trials, cohort studies, case-control studies, cross-sectional studies, and case series were included. Studies published as full-text, abstract only, and unpublished data were included. There were no restrictions by language of publication. For RCTs, studies using standard of care, an alternative intervention, or a placebo treatment as the control group were included. Studies with no control group were also included in the review. Although evidence from RCTs is the gold-standard for assessing effectiveness to establish an intervention for clinical practice, non-RCTs, which are at high risk of bias, were included given the paucity of evidence and to inform future trials.



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**Study Selection**

All potentially eligible records were independently screened by title and abstract by two review authors (RP and SM). The full text of studies were retrieved for each included record, and two review authors (RP and SM) independently screened studies for final inclusion. Reasons for excluding studies were recorded. Any disagreements were resolved through discussion and evaluated by a third review author (PG) if necessary.

**Primary and Secondary Outcomes**

The primary outcome was improvement of swallowing dysfunction as measured by imaging studies, such as videofluoroscopic swallow study (VFSS) and fiberoptic endoscopic evaluation of swallowing (FEES). Secondary outcomes included: 1) swallowing function as reported by a clinician (e.g., a clinical swallowing examination) or by the child and/or their caregiver (e.g. patient reported outcome measures), 2) child's feeding ability (e.g., food type(s) the child can consume, ease of feeding, need for any form of enteral tube feeding, duration of time required for feeding, and child's overall experience of feeding), 3) child's health status (e.g., anthropometric measurements, medical care, and home care), 4) social impact on child (e.g., participation at mealtime), 5) impact on caregiver (e.g., stress associated with feeding), 6) potential adverse events and harms (e.g., aspiration pneumonia and/or recurrent chest infections, mortality), 7) child's quality of life as reported by the child and/or caregiver, and 8) caregiver's quality of life.

**Data Extraction and Management**

One review author (RP) extracted the data and a second review author (SM) verified the extracted data. Data regarding the methods, participants, interventions, outcomes, results was extracted into an a priori



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developed data extraction form. Any disagreements were resolved through discussion, and if necessary, by involving a third review author (PG). Missing data was obtained by contacting study authors by email, up to a maximum of three email attempts.

**Risk of Bias**

Risk of bias for all included studies was assessed independently by two review authors (RP and SM). Any disagreements were resolved through discussion, and if necessary, by involving a third review author (PG). For randomized controlled trial studies, risk of bias was assessed using the Cochrane Collaboration's tool,[26] according to the following domains: random sequence generation, allocation concealment, blinding of participants and researchers, blinding of outcome assessment, incomplete outcome data, selective reporting, and other bias. Each domain was judged as low, high, or unclear risk of bias.[26] For observational studies, risk of bias was assessed using a modified version of the Newcastle-Ottawa assessment tool,[27] according to the following domains: selection, comparability, and outcome. Each domain was then rated using the star system following the coding manual instructions.[27] The scale ranged from 0, indicating the highest risk of bias (i.e., none of the criteria fulfilled), to 7, indicating the lowest risk of bias. The overall risk of bias was summarized across the studies included in the review. The risk of bias for each study was considered when evaluating treatment effects.

**Statistical Analysis**

Randomized controlled trials and observational studies were considered separately. A summary statistic was calculated for the outcomes in each study and grouped according to the pre-defined primary and secondary outcomes. A risk ratio was calculated for dichotomous outcomes. A standardized mean difference was calculated for continuous outcomes, where the numerator was the difference in change scores between the groups and the denominator was the pooled SD of the combined pre and post



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outcome scores across groups.[28] Median (range) was converted to mean (SD) when necessary. Forest plots of effect estimates were constructed for each study and an overall estimate. Given the heterogeneity between the included studies, a meta-analysis was not conducted, and a pooled intervention effect was not calculated. Review Manager (RevMan) was used to conduct the analyses.

**Patient and Public Involvement**

There was no patient or public involvement in this review.

**Results**

Of the 1015 unique records screened by title/abstract, 28 full text records were assessed for eligibility, and 17 records met eligibility criteria for the systematic review. An additional 3 articles were included from reference lists and grey literature. Of the 20 records that met eligibility criteria for systematic review, 10 full text articles were included in systematic review (Figure 1) and 10 abstracts were reported on in Appendix B.

**Study Characteristics**

A total of 5 randomized controlled trials (RCT)[29–33], 4 prospective case series (i.e., before and after study with no control group)[34–37], and 1 retrospective cohort study (with a control group)[38] were included in the review (Table 1). One RCT (Gao et al 2018[30]) included NMES in both treatment and control groups, as well as a co-intervention in the treatment group, and therefore, was only the control group was analyzed. In addition, we were unable to obtain the required data to analyze the study by Christiaanse et al quantitatively,[38] and therefore, this study was analyzed qualitatively. Thus, four studies (4 RCT) were analyzed quantitatively[31–33,39] and 6 studies were analyzed qualitatively.[30,34–38]



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The studies were from 6 countries: Egypt,[33] China,[29,30] Canada,[35] Turkey,[31] South Korea,[32,34] and USA.[36–38] The mean age of the included participants was less than 4 years in 8 studies[29–31,33,35–38] and the mean age was 6 years in one study.[32] One study did not specify the age of the participants.[31] A total of 6 studies included participants exclusively with neurological impairment (e.g., cerebral palsy)[29,30,32–35] and 4 studies included some participants with neurological impairment.[31,36–38] The follow up period ranged from a minimum of 4 weeks in 2 studies[31,34] to a maximum of 8 months in one study.[36] The sample size of the studies ranged from 5 participants[34] to 62 participants per group (control and treatment groups)[30] (Table 1). Most studies used VFSS before and after the intervention to diagnose dysphagia and evaluate response to treatment. Study funding sources are provided in Appendix C.

**Intervention Characteristics**

Most studies placed the electrodes around the hyoid bone and thyroid notch,[29,30,33,35–38] and in one study, electrode placement was further adjusted on an individual basis.[35] One study placed electrodes on the sublingual muscles,[30] and one study placed electrodes on the submental region.[34] The frequency of sessions ranged from 5 days per week[29–31,34] to 1-2 times per week.[32,33,35–38] The session duration ranged from 20 minutes[29,30,32] to 1 hour,[31,36] and treatment duration ranged from 4 weeks[31,34] to 6 months.[37] In most studies, the voltage applied was 80 Hz and current intensity was adjusted until a therapeutic level was reached.[31,32,34–38] A range of interventions were employed for the control groups, but they all included some form of muscle stimulation (e.g., oral motor exercises)[29–33,37,38] and two studies included a placebo/sham-NMES treatment[32,33] (Table 2).

**Outcomes**

*Swallowing function*



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All studies reported on swallowing dysfunction (Table 3 summarizes outcomes reported across studies), and in all cases, swallowing function improved over the course of NMES treatment. However, only five used imaging to define swallowing function, and therefore, we grouped swallowing dysfunction as measured by imaging or clinical exam for this outcome. The standardized mean difference for improvement of swallowing dysfunction in treatment compared to control groups in the RCTs ranged from 0.18 (95% CI -0.7, 1.06)[32] to 1.49 (95% CI 0.57, 2.41)[31] (Figure 2). The study by Lv et al 2019[29] (RCT) exhibited a graded improvement in swallowing function with increasing current intensity in the different treatment groups (Figure 2). In two small prospective case series studies, all participants improved in swallowing function (Supplementary Table 1). In the study by Ma et al 2019[34] (prospective case series study) and Gao et al 2018[30] (RCT treated as cohort study), the scales used to grade swallowing dysfunction showed an overall improvement (Supplementary Table 1). Specifically, in the Gao et al 2018 study, 58% of participants improved their swallowing effectiveness. Improvement in the swallowing activities ranged from 3% improvement on the chewing movement subscale to 11% on the free eating subscale.[30] In the Ma et al 2019 study, there was a 45% improvement in swallowing semisolids and 33% improvement in swallowing liquids as measured by the Penetration Aspiration Scale (PAS).[34] In the Andreoli et al 2019 study, there was a statistically significant improvement in mean swallowing function as measured by the Functional Oral Intake Scale (FOIS) (from 3.07 [1.94] to 4.47 [2.26]). In the Christiaanse et al 2011 study, there was no significant difference observed between the treatment and control groups for change in swallowing function measured by FOIS(Table C).[37]

*Child's feeding ability*

Eight of 10 studies reported on child's feeding ability.[29–32,35–38] In all but one study, there was improvement in feeding ability over the course of the NMES treatment (Figure 2, Table C). Similarly, the study by Lv et al 2019 exhibited a graded improvement in feeding ability with increasing current



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intensity in different treatment groups (Figure 2).[29] Marcus et al 2019 reported that all participants improved in their ability to swallow different consistencies safely and 5 of 5 participants progressed from tube feeding to full or partial oral feeding.[35] Similarly, all participants in the Rice et al 2012 study improved in their ability to swallow different consistencies.[36] There was an improvement in the feeding ability of 69% of participants in the Gao et al 2018 study.[30] In the Andreoli et al 2019 study, improvement was noted in 7 of 8 patients with gastrostomy dependence; for patients without gastrostomy dependence, full oral feeds without restriction was achieved in 6 of 7 patients (Supplementary Table 1).[37] The standardized mean difference for improvement in feeding behaviours compared with the control group in the study by Song et al 2015[32] (RCT) was 0.39 (95% CI -0.49, 1.28). The exception was the study by Christiaanse et al 2011[38] (retrospective cohort study), which found that the intervention was not associated with an improvement in feeding ability compared with the control group (Supplementary Table 1).

*Child's health status*

Lv et al 2019[29] reported on nutritional status, which improved over the course of treatment with standardized mean differences ranging from 0.59 (95% CI -0.15, 1.32) to 1.33 (95% CI 0.53, 2.13) for the different treatment groups compared with the control group using increasing voltage of NMES. El-Sheikah et al 2020[33] found that weight increased over the course of treatment (standardized mean difference of 0.04 (95% CI -0.58, 0.66) (Figure 2).

*Potential adverse events and harms*

Marcus et al 2019[35] reported on adverse events and harms, which included mild skin irritation at the placement of electrode and resolved without treatment (N = 6). Andreoli et al 2019[37] reported that the



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NMES was well tolerated and there were no treatment complications. Adverse events were not reported in the other studies.

*Child's quality of life as reported by the child*

Gao et al 2018[30] reported on child's quality of life using the Gessell scale and there was an overall improvement over the course of treatment. More specifically, there was a mean improvement of 7% for the adaptabilities subscale, 8% for the language behaviour subscale, and 6% for the personal social development subscale (Supplementary Table 1).

*Impact on caregiver*

Marcus et al 2019[35] reported on the impact on the caregiver, specifically 5 of 7 caregivers felt that their child's feeding improved and were satisfied with the intervention. However, a common reported negative aspect was the frequent need to travel to the hospital for treatments.

No studies reported on the social impact on child or caregiver's quality of life (Table 3).

**Quality of Studies**

The risk of bias for RCTs were assessed using the Cochrane collaboration tool.[26] All studies were rated low for attrition bias and reported bias, most studies had a high level of bias for areas related to blinding, most studies were low for random sequence generation, and all studies were unclear for allocation concealment (Supplementary Table 2). Furthermore, for all the RCTs, there was no reported primary outcome or sample size estimation. The observational studies were assessed for risk of bias using a modified version of the Newcastle-Ottawa scale[27]: 3 of 5 studies scored 5 (out of 7),[30,35,36] one study scored 1 (out of 7)[34], one study score 5 (out of 7),[37] and one study scored 6 (out of 8)[38] (Supplementary Table 3).



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## Discussion

### Key Findings

Dysphagia is a serious, often chronic health problem with significant impacts for the child, family, and the healthcare system.[4,5,8,9,40] It can lead to poor oral intake and malnutrition, aspiration and pneumonia, as well as impact daily activities and social interaction with peers.[4,5,40] Parents and caregivers of children with dysphagia experience increased anxiety and stress,[8,9] and there are increased costs and utilization of the healthcare system.[10,11] NMES is a proposed intervention for the treatment of dysphagia. In this review, we analyzed 10 studies, including 5 randomized controlled trials,[29–33] 4 prospective case series studies without a control group,[34–37] and 1 retrospective cohort study[38] where NMES was used to treat dysphagia in children. Overall, both the RCTs and observational studies reported an improvement in swallowing dysfunction following NMES intervention, which demonstrates that NMES treatment may be beneficial for children with dysphagia. However, given the limitations in the primary studies, well-designed trials are needed before adopting this intervention in clinical practice.

### Findings in the Context of Previous Research

This is the first systematic review to investigate the use of NMES for children with dysphagia. Several systematic reviews and meta-analysis have been conducted in different populations of adults with dysphagia (e.g., stroke, head and neck cancer), with some reported benefits on swallowing dysfunction,[20–22] however, these results need to be interpreted with caution in children. Two recent reviews examined the management and interventions for children with dysphagia,[24,41] but no studies on NMES were included. In addition, the population of children with dysphagia is heterogenous, including neonates and infants with a developing muscular system, children with a disordered neuromuscular system (such as children with cerebral palsy), and children with a developed neuromuscular system (such as children with acquired dysphagia).[42]



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To date, the studies in this area are limited in number, and many demonstrated moderate to high risk of bias. For example, most RCTs did not blind participants, personnel, and outcome assessment. Though studies randomized patients to treatment and control groups, there was insufficient information provided about allocation concealment across all studies. In the Song et al study,[32] there was a significant difference between treatment and control groups at baseline. The observational studies had similar issues: the representativeness of the sample was unclear in two studies,[34,38] there was a significant difference between treatment and control groups in one study,[38] and one study provided insufficient data on participants.[34] Lastly, in 7 of 10 studies, the follow up period was less than 6 months.[29–35] A longer follow-up period (e.g. 1 year) is important to determine whether any benefit derived from NMES is temporary or sustained. Together, these factors contribute to moderate to high risk of bias in the included studies.

**Limitations**

This review has several limitations. First, due to clinical heterogeneity, we did not calculate a pooled intervention effect and conduct a meta-analysis. Second, we included a wide range of studies, several of which did not have a control group. In these cases, it cannot be determined that the observed effect was due to NMES. Some of these studies had additional intervention components, such as oral motor exercises, and this may have contributed to the observed effect. In addition, there is a natural maturational development that occurs with time, which may have also contributed to the observed effect. However, when there is limited evidence, it is important to survey the literature more broadly, and therefore, we included these studies in our review. There were 10 abstracts and unpublished studies that were not included in the analysis, however, there was insufficient data reported to include these reports in our review. Third, the populations of children with dysphagia in the included studies are



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clinically heterogenous – neonates and infants, children with neuromuscular disorders, and children with acquired dysphagia. The populations different physiologically, which has implications for use of NMES. Finally, some studies may have been missed despite a comprehensive systematic search of the literature.

**Implications for Future Research**

We have several recommendations for future trials. We recommend the use of imaging studies to objectively evaluate swallowing function, such as VFSS and FEES, which are established imaging techniques used to evaluate swallowing function.[43,44] Furthermore, it is imperative to use validated outcome measures for children with dysphagia that can be applied across different pediatric populations. In terms of validated patient and/or parent reported outcomes, there are several potential measures,[45–48] however, there is no established standard that would allow for systematic reporting and comparison across studies. Second, only several of the a priori primary and secondary outcomes were reported across the studies. Dysphagia has impacts across biological, social, and psychological realms, and therefore, it is important for future trials to assess a broad range of outcomes to develop a more complete clinical picture. For example, none of the studies in this review reported on social impact on the child, impact on the caregiver, and the caregiver's quality of life. In order to ensure outcomes important to patients and their families are assessed, we recommend patients and their families be included as partners when designing future studies. In addition, the effects of the intervention should be assessed in different patient populations (e.g., young infants and older children) and aetiologies (e.g., premature infants, neurologic impairment). It would be important to assess outcomes for a sufficiently long duration of follow up to determine if the effect of NMES persists over time. However, in most of these studies, the follow up period was less than 6 months. It is also important to establish standards for the implementation of NMES regarding probe placement, duration, frequency, voltage, and current



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intensity. Finally, all trials should be reported according to the CONSORT guidelines, including sample size justification as well as blinding participants and personnel, which were notably missing in many of these studies.

**Conclusions**

Dysphagia in childhood has important health impacts for the child and their family, as well as the healthcare system. Evidence from small trials with moderate to high risk of bias and observational studies suggest that NMES improves swallowing function in children. There is limited evidence on other important outcomes. Further high-quality trials are needed before NMES is adopted in routine clinical practice.

**Acknowledgements**

**Competing Interests:** There are no competing interests to disclose.

**Funding/Support:** This study was supported by a grant from the Pediatric Consultants Partnership, The Hospital for Sick Children, Toronto, Canada.

**Role of Funder/Sponsor:** The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

**Authors Contributions:** Ms. Roni Propp conceptualized and designed the study, collected data, carried out the analyses and interpretation of data, and drafted and revised the manuscript. Drs. Peter Gill, Eyal Cohen, Jeremy Friedman and Ms. Sherna Marcus conceptualized and designed the study, interpretated the data, and reviewed and revised the manuscript. Ms. Lily Ren conceptualized and designed the study, designed and executed the search strategy, and reviewed and revised the manuscript. Dr. Sanjay



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Mahant conceptualized and designed the study, collected data, analysed and interpreted the data, drafted the initial manuscript, and reviewed and revised the manuscript. All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

**Data Statement:** Not applicable.

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**Table 1. Study Characteristics**

| Identification | | | Number of participants | | Characteristics of participants | | | | Follow Up Time Length |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Treatment | | Control | | |
| Author | Year | Country | Treatment | Control | **Age**, Years Mean (SD) | **Neurological impairment**, N (%) | **Age,** Years Mean (SD) | **Neurological impairment**, N (%) | |
| **Randomized controlled trials** | | | | | | | | | |
| El-Sheikh[33] | 2020 | Egypt | 20 | 20 | 3.26 (1.16) | 20 (100%) | 3.96 (0.98) | 20 (100%) | 2 months |
| Lv[29] | 2019 | China | 45* | 15 | 3.93 (0.73) | 45 (100%) | 3.92 (0.85) | 15 (100%) | 3 months |
| Gao[30]** | 2018 | China | 62 | 62 | 3.01 (1.12) | 62 (100%) | 3.02 (1.02) | 62 (100%) | 3 months |
| Serel Arslan[31] | 2017 | Turkey | 12 | 12 | 1.01 (0.63) | 1 (8.3%) | 1.28 (0.93) | 1 (8.3%) | 1 month |
| Song[32] | 2015 | South Korea | 10 | 10 | 6.2 (2.78) | 10 (100%) | 6.00 (2.40) | 10 (100%) | 2 months |
| **Prospective case series** | | | | | | | | | |
| Andreoli[37] | 2019 | USA | 15 | --- | 2.51 (3.20) | 11 (73%) | --- | --- | 6 months |
| Ma[34] | 2019 | South Korea | 5 | --- | Not reported | 5 (100%) | --- | --- | 1 month |
| Marcus[35] | 2019 | Canada | 7 | --- | 0.66 (0.37) | 7 (100%) | --- | --- | 2-4 months |
| Rice[36] | 2012 | USA | 5 | --- | 1.45 (1.05) | 2 (40%) | --- | --- | 3-8 months |
| **Retrospective cohort studies** | | | | | | | | | |
| Christiaanse[38] | 2011 | USA | 47 | 46 | 2.58 (1.66) | 32 (68%) | 0.83 (0.68) | 16 (35%) | 6 months |

*3 treatment groups; 15 participants in each group

**Included NMES in both treatment and control groups, and therefore, was analyzed as a prospective case series.



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**Table 2. NMES and Control Group Intervention Details**

| Author | Year | Additional intervention components | Electrode placements | Frequency of sessions | Duration of NMES session | Duration of treatment | Voltage applied | Current intensity | Control Intervention |
|---|---|---|---|---|---|---|---|---|---|
| **Randomized controlled trials** | | | | | | | | | |
| El-Sheikh[33] | 2020 | Oral motor exercises | (1) over the neck between hyoid and jaw (2) side of face | Twice weekly | 20 minutes | 2 months | 60 Hz | 7mA to 15mA | Oral motor exercises; placebo NMES |
| Lv[29] | 2019 | --- | (1) above hyoid bone, (2) above the thyroid upper notch | 5 days a week | 20 minutes | 3 months | 80Hz | 10mA, 15mA, 20mA (3 treatment groups) | Routine rehabilitation training (oral massage and acupuncture) |
| Gao[30]* | 2018 | Oral motor exercises, Acupuncture | Superficial muscles and sublingual muscles | 5 days a week | 20 minutes | 3 months | 20-30 Hz | 3-7 mA | Oral motor exercises, NMES |
| Serel Arslan[31] | 2017 | Thermal tactile application (TTA), Hyolaryngeal mobilization | (1) below the jaw, (2) above the thyroid notch | 5 days a week | 30 minutes | 1 month | 80 Hz | Until a therapeutic level was reached | Non-nutritive stimulations |
| Song[32] | 2015 | Oral sensorimotor treatment | (1) over the throat between jaw and hyoid (2) between hyoid and thyroid notch | Twice weekly | 20 minutes | 2 months | 80 Hz | Typical level ranged from 3mA to 5 mA (determined by palpation for muscle contraction) | Oral sensorimotor treatments and sham-NMES |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Prospective case series studies** | | | | | | | | | |
| Andreoli[37] | 2019 | Comprehensive feeding therapy program | (1) superior aspect of the thyroid cartilage, (2) along tongue base superior to hyoid bone | Weekly | Up to 50 minutes or as tolerated | 6 months | 80 Hz | 7.5 (2.7) mAmp (maximal mean (SD)) | No control group |
| Ma[34] | 2019 | --- | Submental region | 5 days a week | 30 minutes per day | 1 month | 80Hz | Until the patients felt a minimal stimulation level | No control group |
| Marcus[35] | 2019 | Feeding to elicit swallowing during NMES session | (1) above the hyoid, or (2) around the thyroid notch Location determined based on the child's swallowing impairment | Twice a week for 2 months; Further 2 months for those in whom improvement was not observed at 2 months | 40 minutes (median; range 20-45) | 2 to 4 months | 80 Hz | Range from 3 to 16 mA, based on the response of the child | No control group |
| Rice[36] | 2012 | Oral motor stimulation techniques between sessions | (1) above the thyroid notch, or (2) around the thyroid notch | 1-2 times a week | 1 hour | Varied | 80 Hz | 0 to 25mA (adjusted for each child to their highest tolerated level) | No control group |
| Retrospective cohort studies | | | | | | | | | |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Christiaanse[38] | 2011 | Non-nutritive oral motor stimulation or a meal (during NMES); Oral motor exercises between sessions | (1) below the jaw, (2) above the thyroid notch<br><br>In general, infants would only have a pair of electrodes overlying the anterior belly of digastric muscles | 2.9 times a week (mean, range 0.7-4.6) | 30-45 minutes depending on patient age | 2.2 months (mean, range 0.5-6.2 months) | 80Hz | Adjusted until a therapeutic level was reached | Usual diet manipulation, oral motor therapy |

*Included NMES in both treatment and control groups, as well as a co-intervention in the treatment group, and therefore, only the control group was analyzed.

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**Table 3. Summary of Outcomes Reported**

| Outcome | Study | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | **El-Sheikh 2020**[33] | **Andreoli 2019**[37] | **Lv 2019**[29] | **Gao 2018**[30] | **Serel Arslan 2017**[31] | **Song 2015**[32] | **Ma 2019**[34] | **Marcus 2019**[35] | **Rice 2012**[36] | **Christiaanse 2011** |
| Swallowing function | | | | | | | | | | |
| Measured by imaging studies | - | - | - | - | + | - | + | + | + | + |
| Reported by clinician | - | + | + | + | - | + | - | - | - | - |
| Reported by child and/or their caregiver | + | - | - | - | - | - | - | - | - | - |
| Child's feeding ability | - | + | + | + | + | + | - | + | + | + |
| Child's health status | + | - | + | - | - | - | - | - | - | - |
| Social impact on child | - | - | - | - | - | - | - | - | - | - |
| Impact on caregiver | - | - | - | - | - | - | - | + | - | - |
| Potential adverse events and harms | - | - | - | - | - | - | - | + | - | - |
| Child's quality of life | - | - | - | + | - | - | - | - | - | - |
| Caregiver's quality of life | - | - | - | - | - | - | - | - | - | - |

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**Figure Legends**

**Figure 1. PRISMA Flowchart**

**Figure 2. Outcomes in Randomized Controlled Trials.**[1] [1]Studies did not report all outcomes; for standardized mean difference, point estimate of 0 = no effect, and for risk ratio, point estimate of 1 = no effect. Abbreviations, RCT, randomized controlled trial.

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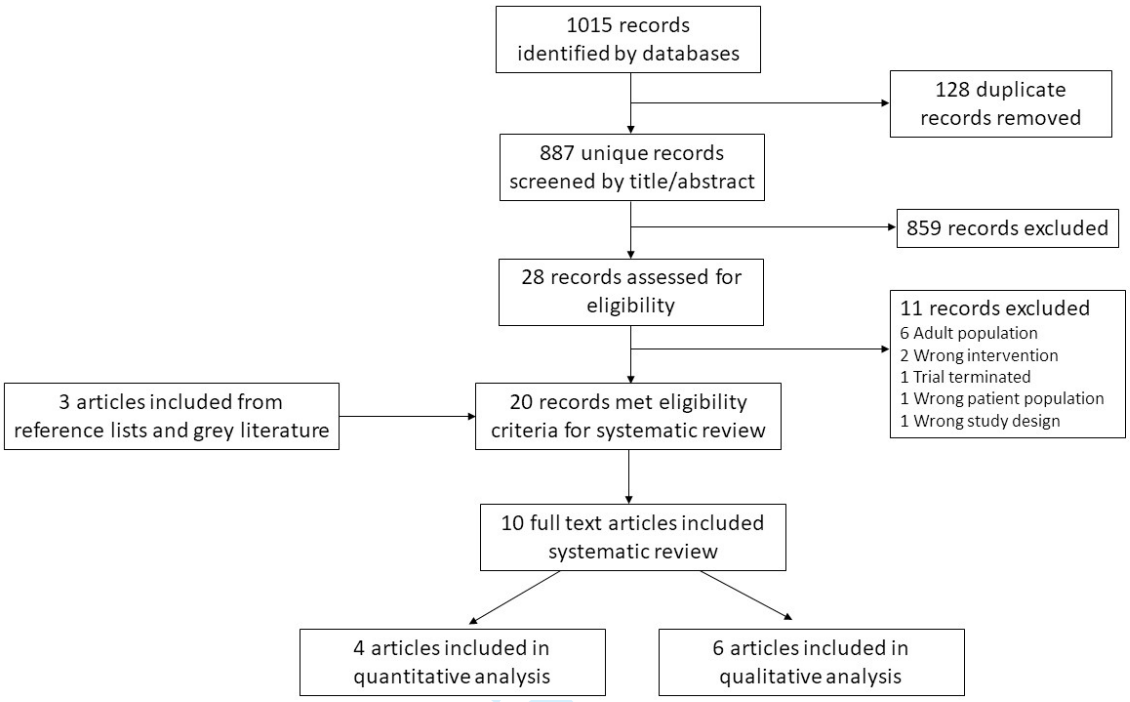
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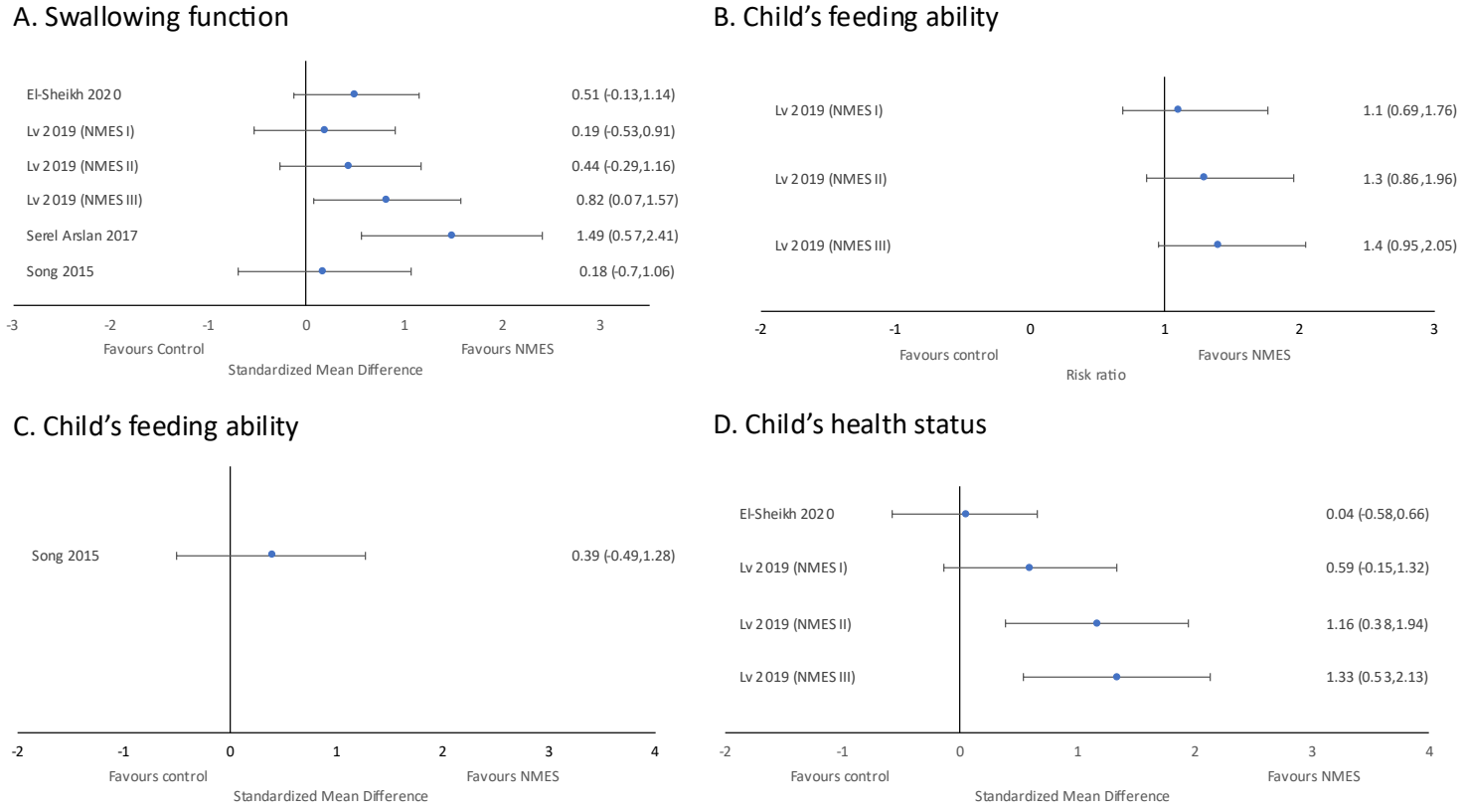
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**Figure 1. PRISMA Flowchart**




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**Figure 2. Outcomes in Randomized Controlled Trials**[1]




[1]Studies did not report all outcomes; for standardized mean difference, point estimate of 0 = no effect, and for risk ratio, point estimate of 1 = no effect. Abbreviations, RCT, randomized controlled trial.

## Supplementary Tables

### Supplementary Table 1. Outcomes in Observational Studies[1]

| Study | Number of Participants | Outcome | Scale | Description |
|---|---|---|---|---|
| Andreoli 2019 | 15 | Swallowing function<br><br>Feeding ability | Functional Oral Intake Scale[2]<br><br>Number of participants | (Mean (SD)): Pre = 3.07 (1.94), Post = 4.47 (2.26)<br>Of those with aspiration and penetration 10/12 improved and of those with aspiration 7/10 resolved<br>For patients with gastrostomy dependence, improvement was noted in 7/8 patients; for patients without gastrostomy dependence, full oral feeds without restriction was achieved in 6/7 patients |
| Christiaanse 2011[36] | 93 | Swallowing function<br><br>Feeding ability | Functional Oral Intake Scale[2]<br>Number of participants | Mean change (SD): Treatment = 0.77 (1.22), Control = 0.39 (0.98), however, this difference was not statistically significant<br>Treatment group: 80% of patients were fed by the same method before and after the intervention; 4 participants were no longer using enteral tubes at 6 month follow-up<br>Control group: 70% of patients were fed by the same method before and after the intervention; 7 participants were no longer using oral tubes at 6 month follow-up |
| Ma 2019[32] | 5 | Swallowing function | Penetration Aspiration Scale[3] | Semisolid (Mean (SD)): Pre = 3.8 (1.5), Post = 2.1 (1.2)<br>Liquid (Mean (SD)): Pre = 6.4 (2.1), Post = 4.3 (2.5) |
| Marcus 2019[33] | 7 | Swallowing function<br><br>Child's feeding ability<br><br>Impact on caregiver | Number of participants<br>Number of participants<br><br>Number of participants | All participants improved<br><br>5 of 7 participants improved<br>Pre: 5 not safe for oral feeding (tube fed)<br>Post: 3 full oral feeding, 2 partial oral feeding, 0 tube fed<br>5 of 7 caregivers felt feeding improved and were satisfied with the intervention; frequent need to travel to the hospital for treatments |
| Gao 2018[28] | 62[4] | Child's feeding ability<br><br><br>Child's quality of life | TDS Classification<br><br>Efficacy<br><br>Gessell | 36 of 62 participants improved<br><br>43 of 62 participants had an effective or significantly effective response to intervention<br>Adaptabilities: 7% improvement |

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| | | | | |
|---|---|---|---|---|
| | | Swallowing function | Number of participants[5] | Language behaviour: 8% improvement<br>Personal social development: 6% improvement<br>Movement of lips and tongue: 6 of 62 participants improved<br>Chewing movement: 2 of 62 participants improved<br>Swallowing function: 5 of 62 participants improved<br>Free eating: 7 of 62 participants improved |
| Rice 2012[34] | 5 | Swallowing function<br><br>Child's quality of life | Number of participants<br><br>Number of participants | All participants improved; Improved swallow safety and ability to advance in consistencies<br>All participants improved |

[1]Only studies without a control group were included in this table
[2]Higher scores indicate improvement
[3]Lower scores indicates improvement
[4]Number of participants included in our analysis (control group only)
[5]Number of participants in denominator is participants who exhibited this activity before treatment

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**Supplementary Table 2. Risk of Bias Assessment for the Included Observational Studies Using the Newcastle Ottawa Scale (NOS)[1]**

| Author | Year | Control Group | Selection (3/4 items)[2] | Comparability (1 item) | Outcome (3 items) | Score (Max = 7/8)[3] |
|---|---|---|---|---|---|---|
| Andreoli[35] | 2019 | No | ★ ★ ★ | ★ | ★★ | 5 |
| Ma[32] | 2019 | No | ★ | | | 1 |
| Marcus[33] | 2019 | No | ★ ★ ★ | ★ | | 4 |
| Gao[28 4] | 2018 | No | ★ ★ | ★ | ★ | 4 |
| Rice[34] | 2012 | No | ★ ★ | ★ | ★ | 4 |
| Christiaanse[36] | 2011 | Yes | ★ ★ ★ ★ | | ★ ★ | 6 |

[1]A modified version of the Newcastle Ottawa scale was used (Appendix C).

[2]For prospective case series (with no control group), selection consists of 3 items. For retrospective studies (with control group), selection consists of 4 items.

[3]For prospective case series (with no control group), scores ranged from 0 (highest risk of bias) to 7 (lowest risk of bias). For retrospective studies (with control group), scores ranged from 0 (highest risk of bias) to 8 (lowest risk of bias).

[4]This study included NMES in both treatment and control groups, and therefore, was analyzed as a prospective case series.

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**Supplementary Table 3. Risk of Bias Assessment for the Included Randomized Controlled Trials Using the Cochrane Collaboration Tool**[1]

| | | Selection bias | | Performance bias | | Detection bias | Attrition bias | Reporting bias | Other bias | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Author** | **Year** | Random sequence generation | Allocation concealment | Blinding of participants | Blinding of personnel | Blinding of outcome assessment | Incomplete outcome data | Selective reporting | Similarity at baseline | Follow up |
| El-Sheikh[31] | 2020 | ? | ? | + | ? | ? | ? | + | + | - |
| Lv[27] | 2019 | + | ? | - | - | - | + | + | + | - |
| Serel Arslan[29] | 2017 | + | ? | - | - | ? | + | + | - | - |
| Song[30] | 2015 | ? | ? | + | - | - | + | + | + | - |

[1] + indicates low risk of bias, - indicates high risk of bias, ? indicates unclear.

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**Appendix A. Search Strategies for electronic databases**

**MEDLINE (Ovid) search strategy**

1. deglutition disorders/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw,kf.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw,kf.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw,kf.
5. (feeding* or regurgit*).tw,kf.
6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw,kf.
7. or/1-6
8. oropharynx/ or oropharyn*.tw,kf.
9. (dysphag* or disorder* or dysfunc* or impair*).tw,kf.
10. 8 and 9
11. 7 or 10
12. electric stimulation therapy/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw,kf.
14. 12 or 13
15. exp child/ or exp "congenital, hereditary, and neonatal diseases and abnormalities"/ or exp infant/ or adolescent/ or exp pediatrics/ or child, abandoned/ or exp child, exceptional/ or child, orphaned/ or child, unwanted/ or minors/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre-term or preterm* or premature birth* or NICU or preschool* or pre-school* or kindergarten* or kindergarden* or elementary school* or nursery school* or (day care* not adult*) or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or pre-pubesc* or prepubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

**Embase (Ovid) search strategy**

1. dysphagia/ or swallowing/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw,kw.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw,kw.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw,kw.
5. (feeding* or regurgit*).tw,kw.
6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw,kw.
7. or/1-6
8. oropharynx/ or oropharyn*.tw,kw.
9. (dysphag* or disorder* or dysfunc* or impair*).tw,kw.
10. 8 and 9
11. 7 or 10
12. electrotherapy/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw,kw.

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14. 12 or 13
15. juvenile/ or exp adolescent/ or exp child/ or exp postnatal development/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre-term or preterm* or premature birth* or NICU or preschool* or pre-school* or kindergarten* or kindergarden* or elementary school* or nursery school* or (day care* not adult*) or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or pre-pubesc* or prepubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

**PsycINFO (Ovid) search strategy**

1. dysphagia/ or swallowing/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw,id.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw,id.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw,id.
5. (feeding* or regurgit*).tw,id.
6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw,id.
7. or/1-6
8. oropharyn*.tw,id.
9. (dysphag* or disorder* or dysfunc* or impair*).tw,id.
10. 8 and 9
11. 7 or 10
12. electrical stimulation/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw,id.
14. 12 or 13
15. adolescent development/ or childhood development/ or pediatrics/ or exp Congenital Disorders/ or child characteristics/ or child abuse/ or exp child welfare/ or chronically ill children/ or child neglect/ or child psychiatry/ or child psychopathology/ or exp child care/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre term or preterm* or premature birth or NICU or preschool* or pre school* or kindergarten* or elementary school* or nursery school* or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or prepubesc* or pre pubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

**Cochrane Central Register of Controlled Trials (OVID) search strategy**

1. deglutition disorders/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw.
5. (feeding* or regurgit*).tw.

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6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw.
7. or/1-6
8. oropharynx/ or oropharyn*.tw.
9. (dysphag* or disorder* or dysfunc* or impair*).tw.
10. 8 and 9
11. 7 or 10
12. electric stimulation therapy/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw.
14. 12 or 13
15. exp child/ or exp "congenital, hereditary, and neonatal diseases and abnormalities"/ or exp infant/ or adolescent/ or exp pediatrics/ or child, abandoned/ or exp child, exceptional/ or child, orphaned/ or child, unwanted/ or minors/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre-term or preterm* or premature birth* or NICU or preschool* or pre-school* or kindergarten* or kindergarden* or elementary school* or nursery school* or (day care* not adult*) or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or pre-pubesc* or prepubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

**CINAHL (EBSCO) search strategy**

1. (MH "deglutition disorders") OR TX (dysphagia* OR deglutition OR swallow OR swallows OR swallowing OR feeding* OR odynophagia* OR feeding* OR regurgit*) OR TX (deglut* N3 (abnormal* OR disorder* OR dysfunc* OR impair*)) OR TX (oropharynx* OR trachea* OR lung* OR pulmon* ) N3 aspirat* ) ) OR TX (pharyng* N3 (dysphag* OR dysfunct* OR disorder* OR impair*))
2. ((MH "oropharynx") OR TX (oropharyn*)) AND TX (dysphag* OR disorder* OR dysfunc* OR impair*)
3. 1 or 2
4. (MH "electric stimulation therapy")
5. TX (electrostimulation or NMES or "electrical stimulation" or "electric stimulation" or electrotherapy or "electro therapy")
6. 4 or 5
7. (MH "Child+") OR (MH "Child Behavior+") OR (MH "Child Abuse, Sexual") OR (MH "Child Behavior Disorders+") OR (MH "Child, Medically Fragile") OR (MH "Child Day Care") OR (MH "Child Passenger Safety") OR (MH "Child Abuse+") OR (MH "Child Custody") OR (MH "Child Behavior Checklist") OR (MH "Child Nutritional Physiology+")
8. (MH "Child Development Disorders, Pervasive+") or (MH "Adolescence+") or (MH "Minors (Legal)") or (MH "Child Mortality")
9. SO (child* or pediatric* or paediatric* or adolescent)
10. TX (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or "pre-term" or preterm or "premature birth*" or NICU or preschool* or "preschool*"

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or kindergarten* or "elementary school*" or "nursery school*" or schoolchild* or toddler* or boy or boys or girl* or "middle school*" or pubescen* or juvenile* or teen* or youth* or "high school*" or adolesc* or prepubesc* or "pre-pubesc*")

11. or/7-10
12. 3 and 6 and 11

**Scopus search strategy**

( ( TITLE-ABS-KEY ( dysphagia* OR deglutition OR swallow OR swallows OR swallowing OR feeding* OR odynophagia* OR feeding* OR regurgit* ) ) OR ( TITLE-ABS-KEY ( deglut* W/3 ( abnormal* OR disorder* OR dysfunc* OR impair* ) ) ) OR ( TITLE-ABS-KEY ( ( oropharynx* OR trachea* OR lung* OR pulmon* ) W/3 aspirat* ) ) OR ( TITLE-ABS-KEY ( pharyng* W/3 ( dysphag* OR dysfunct* OR disorder* OR impair* ) ) ) OR ( TITLE-ABS-KEY ( oropharyn* ) AND TITLE-ABS-KEY ( dysphag* OR disorder* OR dysfunc* OR impair* ) ) ) AND ( ( TITLE-ABS-KEY ( electrostimulation OR nmes OR "electrical stimulation" OR "electric stimulation" OR electrotherapy OR "electro therapy" ) ) AND ( TITLE-ABS-KEY ( pediatric* OR paediatric* OR child* OR newborn* OR congenital* OR infan* OR baby OR babies OR neonat* OR "pre term" OR preterm* OR "premature birth" OR nicu OR preschool* OR "pre school*" OR kindergarten* OR "elementary school*" OR "nursery school*" OR schoolchild* OR toddler* OR boy OR boys OR girl* OR "middle school*" OR pubescen* OR juvenile* OR teen* OR youth* OR "high school*" OR adolesc* OR prepubesc* OR "pre pubesc*" ) ) )

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**Appendix B. Citation information for unpublished studies and abstracts**

| Author | Year | Title | Journal |
|---|---|---|---|
| Biernacka | 2019 | Evaluation of physical and emotional well-being of children undergoing electrostimulation of swallowing | J Pediatr Gastroenterol Nutr |
| Winnicka | 2019 | Improving swallowing function due to electrical stimulation in group of children-preliminary results from an innovative method of treatment | J Pediatr Gastroenterol Nutr |
| Winnicka | 2019 | Improvement of oral feeding due to electrical stimulation in children with primary dysphagia | J Pediatr Gastroenterol Nutr |
| Radika | 2018 | Role of NMES in dysphagia management for children with hypertonia and dystonia: a case study | Dysphagia |
| Biernacka | 2017 | Evaluation of physical and emotional comfort of children undergoing electrostimulation of swallowing- Initial report | J Pediatr Gastroenterol Nutr |
| Andreoli | 2016 | Neuromuscular stimulation and feeding in medically complex children | Otolaryngol Head and Neck Surg |
| Demir | 2015 | Swallowing function of a pierre robin sequence with absence epiglottis: A case report | Dysphagia |
| Mitchell | 2010 | The effectiveness of NMES (VitalStim) therapy in the neonatal population | Unpublished |
| Christiaanse | 2005 | Experience with pharyngeal electrical stimulation: A new treatment modality for the management of pediatric dysphagia | Dev Med Child Neurol |
| Chatburn | 2000 | Electrical stimulation for treatment of dysphagia in children failing conventional therapy | Respiratory Care |

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**Appendix C. Study Funding Source**

| Study | Year | Funding Source |
|---|---|---|
| Andreoli[35] | 2019 | Not indicated |
| Christiaanse[36] | 2011 | No financial support |
| El-Sheikh[31] | 2020 | Not indicated |
| Gao[28] | 2018 | Not indicated |
| Lv[27] | 2019 | National Natural Science Foundation of China Early Career Fund, Young Talents of Science, Education and Health in Jiangsu Province |
| Ma[32] | 2019 | Not indicated |
| Marcus[33] | 2019 | Peer-reviewed grant from the Department of Paediatrics, The Hospital for Sick Children, Toronto |
| Rice[34] | 2012 | Not indicated |
| Serel Arslan[29] | 2017 | No financial support |
| Song[30] | 2015 | No financial support |

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | 4-5 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 5 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | 6 |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 6-7 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 6 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | 6 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 6-7 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 6 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 6-9 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 8 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | 9 |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | 9 |

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 8 |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | 9 |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | 10 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | 10, Table 1 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | 14, Online table 5 and 6 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | 11-15, Figure 2 |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | NA |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 14, Online table 5 and 6 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | NA |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 15 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 16 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 17 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 17-18 |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: www.prisma-statement.org.



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